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1 Variances

Theorem. Let X and Y be independent random variables with finite variance, then $\text{Var}[X] + \text{Var}[Y] = \text{Var}[X + Y]$

Proof.

$$\begin{aligned}
 \text{Var}(X + Y) &= -\left(E(X + Y)^2\right) + E\left((X + Y)^2\right) \\
 &= -(E(X) + E(Y))^2 + E(X^2 + 2XY + Y^2) \\
 &= -E(X)^2 - E(Y)^2 - 2E(X)E(Y) + E(X^2) + 2E(XY) + E(Y^2) \\
 &= \text{Var}(X) + \text{Var}(Y) + 2(-E(X)E(Y) + E(XY))
 \end{aligned}$$

So it remains to show that the quantity $E(XY) - E(X)E(Y) = 0$ when X and Y are independent. In fact, this quantity, which is easily shown to be equal to $E((X - E(X))(Y - E(Y)))$ by expanding, is called Covariance ($\text{Cov}(X, Y)$)

When X and Y are independent, then for each possible value of X , $E(Y - E(Y)) = 0$. Since the expectation when there are a bunch of possible outcomes is intuitively equal to the sum of the expected value from those outcomes times the probability of those outcomes, we can apply this to the possible values of X , so the quantity

$$E((X - E(X))(Y - E(Y)))$$

becomes a weighted sum of terms like

$$E((x - E(X))P(X = x)(Y - E(Y)))$$

which is a bunch of terms like

$$E(\text{constant}(Y - E(Y)))$$

which is 0 because $Y - E(Y)$ is always 0 regardless of the value of X due to the fact that X and Y are independent.

An important idea is that independence allows you to take (for independent variables)

$$E(X)E(Y) = E(XY)$$

□

A more formal treatment of independence is given in Level 6.2.

2 The normal distribution

2.1 Density and variance

Theorem. The function

$$\frac{1}{\sigma\sqrt{2\pi}} e^{-\left(\frac{(x-\mu)^2}{2\sigma^2}\right)}$$

is a valid probability density function with mean μ and variance σ^2 and $\sigma > 0$

Proof. So we have a probability distribution based on e^{-x^2} , this is an assumption. We change this to $e^{-\frac{x^2}{2}}$ since it turns out (we will prove this after) that this will make the variance equal 1. Then we have to rescale by a factor, which turns out to be $\sqrt{2\pi}$ (This is what we will prove, but really we just need to know that the area exists which is easy to prove, however the square root of pi is so fun that I can't just not show the proof) so that the area under the curve equals 1.

We will prove that the total area under e^{-x^2} is $\sqrt{\pi}$ then it is clear that by shifting the mean the area will not change and that if we divide x by a constant σ then we are stretching the curve and therefore its area by a factor of σ so we need to divide the correction factor by σ to compensate and keep the area under the curve at 1.

Now, we will consider what happens if we take the e^{-x^2} curve and rotate it about the y axis, like this:

Figure 1 shows a 2 dimensional normal plotted on 3D graph

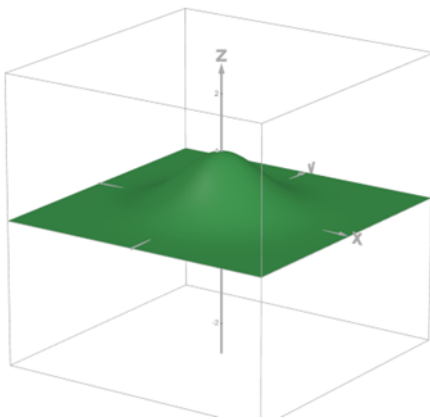


Figure 1

As you can see, it now essentially becomes a function of the distance from the origin d, i.e. $z = e^{-d^2}$. Now by pythagoras this becomes $z = e^{-(x^2+y^2)}$. We want to show that the volume of this surface is π , because then we know that if the value $\int_{-\infty}^{\infty} e^{-x^2} dx$ is k then the value of, the strip of this surface for a certain y value with a small width dy (As illustrated in the figure below) is approximately $(\int_{x=-\infty}^{\infty} e^{-(y^2+x^2)} dx) dy = (e^{-y^2} \int_{-\infty}^{\infty} e^{-x^2} dx) dy = (e^{-y^2} k) dy$,

The justification for the above is in level 6.2.

Figure 2 shows a thin slice of the 2d normal in the above image

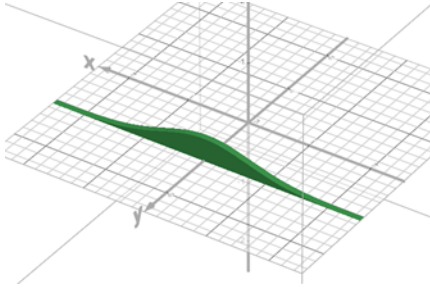


Figure 2

Then the volume under the surface is the sum of the volumes of these tiny strips (as dy gets smaller), ie, as we have seen in the interpretation of the integral as a sum, $\int_{-\infty}^{\infty} (e^{-y^2} k) dy = k^2$. Therefore, if we can show that the volume under the surface which is k^2 is π , then $k = \sqrt{\pi}$ as required.

The normal distribution is very special in the sense that if we take x and y independently normally distributed (which makes the probability density become the product of the probability density for x and y), the resulting distribution has rotational symmetry and depends only on the instance, as it becomes $e^{-(x^2+y^2)}$, which is a function of the distance by pythagoras.

Now, to prove that the volume is indeed π , we will consider what happens if we split the volume into concentric rings, like this:

Figure 3 shows a 2d normal on 3d graph approximated by concentric rings of the right height

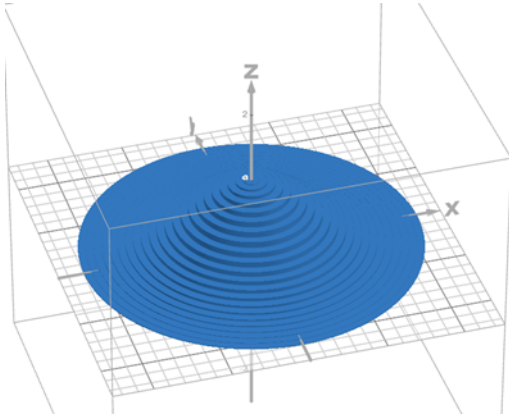


Figure 3

Suppose the width of the rings is dr and their distance from the origin is r , then the height of the rings is e^{-r^2} and the circumference of them is $2\pi r$ so the volume of each ring is $2\pi r e^{-r^2} dr$. So, the total volume under the figure is $\int_0^{\infty} 2\pi r e^{-r^2} dr$, again by the interpretation of integration as a sum. Note that you can check by differentiating that an antiderivative of $2\pi r e^{-r^2}$ is $-\pi e^{-r^2}$. So, we have to evaluate $\left[-\pi e^{-r^2} \right]_0^{\infty}$ which is $0 - [-\pi]$ which is π as required.

Comment: This is often the case in mathematics, the idea seeing things in 2 different ways is extremely common (like here with strips vs rings).

Variance:

If we have a standard normal $\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{x^2}{2}} dx$ then the variance is given by $E(x^2)$ since the mean is 0 by symmetry so we need to work out $\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} x^2 e^{-\frac{x^2}{2}} dx$ and show that it is indeed 1, then it becomes clear that the variance is always σ^2 due to scaling properties of variances (ie, if we were to stretch the normal distribution by $2x$, we get this effect by multiplying σ by 2 in the formula and the variance multiplies by 4 by variance scaling properties and the variance is unchanged by shift in the mean since it is related to distance from mean)

Now, we try integrating $\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{x^2}{2}} dx$ (which, keep in mind, is 1) by parts with $e^{-\frac{x^2}{2}}$ being the part to differentiate

and 1 being the part to integrate. This will give us $\frac{1}{\sqrt{2\pi}} \left[\left[x e^{-\frac{x^2}{2}} \right]_{-\infty}^{\infty} + \int_{-\infty}^{\infty} x^2 e^{-\frac{x^2}{2}} dx \right]$ (the $-$ in the integration by parts formula cancels with the $-$ from differentiating the exponential). Now, since $x e^{-\frac{x^2}{2}}$ goes to 0 as x goes to positive and negative infinity (This is because clearly the derivative of a normal distribution must go to 0 on either side and this is just minus that, or alternatively because the exponential term decays much faster than x grows), this formula, which is equal to 1, reduces to the integral $\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} x^2 e^{-\frac{x^2}{2}} dx$ that we needed to find.

□

Remark. Recall that in Level 5.1 I gave as an example in the “reduction formulae” subsection that we can show that $\int_0^{\infty} x^n e^{-x} dx$ is a generalization of $n!$. Although we will not see here why this is the “canonical” extension, I will note that the value of $(\frac{1}{2})!$ is... interesting, and quite funny.

We want to compute $\int_0^{\infty} \sqrt{x} e^{-x} dx$. We will set $u = \sqrt{x}$, $du = \frac{1}{2\sqrt{x}} dx$, I say $=$ to mean “can be replaced with in the integral”.

We now have to compute $\int_0^{\infty} 2x e^{-u^2} du = \int_0^{\infty} 2u^2 e^{-u^2} du$. Note that the limits don’t change when we do the substitution in this case. But we just calculated that

$$\int_{-\infty}^{\infty} x^2 e^{-\frac{x^2}{2}} dx = \sqrt{2\pi}$$

which implies that, by setting $u = \frac{x}{\sqrt{2}}$ so that $dx = \sqrt{2} du$

$$\int_{-\infty}^{\infty} x^2 e^{-\frac{x^2}{2}} \sqrt{2} du = \sqrt{2\pi}$$

$$\int_{-\infty}^{\infty} 2u^2 e^{-u^2} \sqrt{2} du = \sqrt{2\pi}$$

$$\int_{-\infty}^{\infty} 2u^2 e^{-u^2} du = \sqrt{\pi}$$

Hence by symmetry,

$$\int_0^{\infty} 2u^2 e^{-u^2} du = \frac{\sqrt{\pi}}{2}$$

Thus, $(\frac{1}{2})! = \frac{\sqrt{\pi}}{2}$. It is extraordinary that π comes up in this context and that it is *square rooted*.

2.2 Sums of normals

Theorem. The sum of two independent normal distributions is a normal distribution. This is “evidence” for the central limit theorem in the sense that if there is a universal limiting distribution it should have this property.

Proof. Note that this is not a fully formal proof but the visual reason why is very pretty so this will be our justification of this fact.

In my proof of the area under a normal distribution curve earlier, I briefly discussed the idea that the normal distribution is special since it is completely rotationally symmetrical if you plot in a higher dimensional space the probability density function of different standard normals. Suppose X and Y are $N(0,1)$ random variables that are independent, then it suffices to show that $aX + bY = N(0, a^2 + b^2)$ since if X and Y were shifted by constants the sum would just shift accordingly and it would all be fine. Here is the joint probability density function of X and Y visualised:

Figure 4 shows another 2d normal plotted on a 3d graph

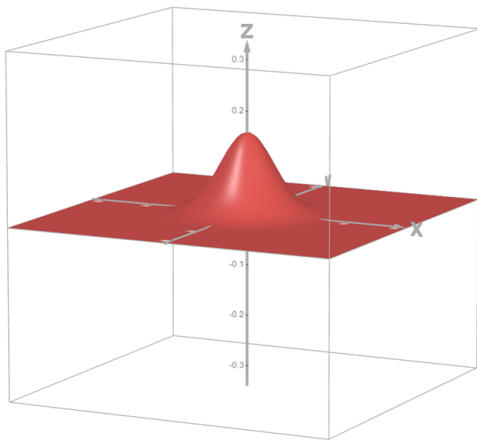


Figure 4

As you can see, there is rotational symmetry. We actually just need to show that $aX+bY$ is normal then we will know from mean and variance additive properties the desired result. Suppose, for example, we want to find the probability density function of $P(3x+2y)$, then the probability density that $P(3x+2y=k)$, intuitively, corresponds to the area under the slice of the above diagram corresponding to $3x+2y=k$. In the end, we will have to rescale the probability density function of $3x+2y$ so that the area under that curve is 1, but it's quite obvious we have proportionality, which is what we actually need. Below is the slices I mean for some values of k .

Figure 5 shows the same 2d normal with parallel vertical planes on the same graph

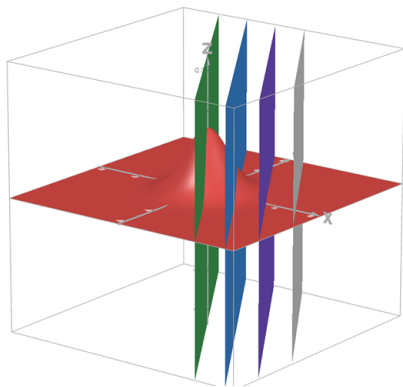


Figure 5

Visually you can see from the rotational symmetry of the image that the function of the area under the red in each of these planes, which as discussed is what we need, is a normal distribution, completing what is not really a proof but rather a visual argument.

□

3 Correlation and regression

3.1 Residual sums of squares

Theorem. Given a finite data set, and a linear regression line $y = a + bx$, the parameters minimizing the sum of squared vertical residuals are given by

$$b = \frac{s_{xy}}{s_{xx}} = \frac{\sum (X_i - \bar{X}) (Y_i - \bar{Y})}{\sum [(X_i - \bar{X})^2]}$$

and

$$a = \bar{y} - b\bar{x}$$

provided $s_{xx} \neq 0$, i.e. the points do not lie all on a vertical line

Proof. The sum of squares as a function of a and b is defined as $\text{RSS}(a, b) := \sum_{i=1}^n (y_i - a - bx_i)^2$.

$$\begin{aligned}\text{RSS}(a, b) &= \sum_{i=1}^n ((y_i - bx_i - \bar{y} + b\bar{x}) + (\bar{y} - b\bar{x} - a))^2 \\ &= \sum_{i=1}^n (y_i - bx_i - \bar{y} + b\bar{x})^2 + 2(\bar{y} - b\bar{x} - a) \sum_{i=1}^n (y_i - bx_i - \bar{y} + b\bar{x}) + \sum_{i=1}^n (\bar{y} - b\bar{x} - a)^2 \\ &= \sum_{i=1}^n (y_i - bx_i - \bar{y} + b\bar{x})^2 + n(\bar{y} - b\bar{x} - a)^2\end{aligned}$$

For each given b , this expression is minimized when the squared bracket is 0 and thus $a = \bar{y} - b\bar{x}$

Now we substitute this a back into our formula and it simplifies to

$$\text{RSS}(b) = \sum_{i=1}^n (y_i - bx_i - \bar{y} + b\bar{x})^2$$

We want to minimize this with respect to b . We expand it out as

$$\begin{aligned}\text{RSS}(b) &= \sum_{i=1}^n (y_i - \bar{y})^2 - 2b \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) + b^2 \sum_{i=1}^n (x_i - \bar{x})^2 \\ &= S_{yy} - 2bS_{xy} + b^2S_{xx} \\ &= S_{yy} + S_{xx} \left(b^2 - 2b \frac{S_{xy}}{S_{xx}} \right)\end{aligned}$$

This is ok since $S_{xx} \neq 0$. This is a quadratic in b that by completing the square is minimized when $b = \frac{S_{xy}}{S_{xx}}$.

□

Proposition. The residual sum of squares from the regression line above is given by $S_{yy}(1 - r^2)$ where r is the PMCC

Proof. Using the formula we proved above, we get

$$\begin{aligned}\text{RSS} &= \sum_{i=1}^n (y_i - (a + bx_i))^2 = \sum_{i=1}^n (y_i - bx_i - \bar{y} + b\bar{x})^2 \\ &= \sum_{i=1}^n (b(\bar{x} - x_i) - (\bar{y} - y_i))^2 = b^2S_{xx} - 2bS_{xy} + S_{yy} = S_{yy} - bS_{xy}\end{aligned}$$

The last equality is by the form we proved of b .

$$= S_{yy} \left(1 - \frac{S_{xy}^2}{S_{xx}S_{yy}} \right) = S_{yy}(1 - r^2)$$

Again since $b = \frac{S_{xy}}{S_{xx}}$.

□

3.2 Correlation coefficients

Theorem. The product moment correlation coefficient is always between -1 and 1, when it is defined (i.e. when $S_{xx}S_{yy} \neq 0$)

Proof. I love this one because it was one of the first times I saw different areas of maths come together.

Let $\vec{a} = (x_1 - \bar{x}, x_2 - \bar{x}, \dots, x_n - \bar{x})$ and $\vec{b} = (y_1 - \bar{y}, y_2 - \bar{y}, \dots, y_n - \bar{y})$. Then the formula for the PMCC is just $\frac{a \cdot b}{|\vec{a}| |\vec{b}|}$. But $a \cdot b = |\vec{a}| |\vec{b}| \cos(\theta)$ so we get $\cos(\theta)$ which is clearly between -1 and 1. Also it is -1 or 1 exactly when all pairs lie on one non-horizontal, non-vertical straight line with the sign depending on the sign of its slope.

□

Corollary. PMCC actually measures (linear) correlation.

Proof. I mean, if you look at the proof above, you see that it's literally the high dimensional cosine of the angle between your vector of x deviations from the mean and y deviations from the mean! It should then be obvious that if the deviations are close together, the cosine of the angle between them should be high, and vice versa, and furthermore the property should hold that scaling both of them shouldn't change the angle between them, which takes care of that part. It also follows that PMCC is only -1 or 1 when x against y makes a straight line, as the deviation from the sample means of x and y in that case will always be the same, just scaled by a constant, thus the cosine of the angle between them will be -1 or 1.

□

Proposition. If we have ranked data x and y with no ties then the PMCC between the ranks of x and y is given by $1 - \frac{6 \sum d^2}{n(n^2-1)}$ where d is the differences between the ranks.

Proof. x_i and y_i are integers 1 to n in any order. Note that

- $r = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}}$ (This is what we want to find).
- $\bar{y} = \frac{n+1}{2}$
- $\bar{x} = \frac{n+1}{2}$

Now

$$\begin{aligned} S_{xx} &= \sum (X_i - \bar{X})^2 = \sum_{i=1}^n \left(i - \frac{n+1}{2} \right)^2 \\ &= \sum_{i=1}^n \left[i^2 - i(n+1) + \frac{(n+1)^2}{4} \right] \\ &= \frac{1}{6} n(n+1)(2n+1) - (n+1) \frac{n(n+1)}{2} + \frac{n(n+1)^2}{4} \\ &= n(n+1) \left[\frac{1}{6} (2n+1) - \frac{n+1}{2} + \frac{n+1}{4} \right] = n(n+1) \left[\frac{1}{12} n - \frac{1}{12} \right] = \frac{1}{12} n(n^2-1) \end{aligned}$$

Note that since x_i and y_i are the same but just reordered, it follows that

$$S_{xx} = \sum (X_i - \bar{X})^2 = \sum (Y_i - \bar{Y})^2 = S_{yy}$$

Therefore $\sqrt{S_{xx} S_{yy}} = \frac{1}{12} n(n^2-1)$. Therefore $r = \frac{12 S_{xy}}{n(n^2-1)}$. Now,

$$\begin{aligned} S_{xy} &= \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}) = \sum_{i=1}^n \left(X_i - \frac{n+1}{2} \right) \left(Y_i - \frac{n+1}{2} \right) \\ &= \sum_{i=1}^n \left[X_i Y_i - \frac{n+1}{2} X_i - \frac{n+1}{2} Y_i + \frac{(n+1)^2}{4} \right] \end{aligned}$$

$$= \sum_{i=1}^n X_i Y_i - \frac{n+1}{2} \sum_{i=1}^n X_i - \frac{n+1}{2} \sum_{i=1}^n Y_i + \sum_{i=1}^n \frac{(n+1)^2}{4}$$

Note that $\sum_{i=1}^n X_i$ and $\sum_{i=1}^n Y_i$ are both $\frac{n(n+1)}{2}$ and so

$$\begin{aligned} &= \sum_{i=1}^n X_i Y_i - \frac{n+1}{2} \frac{n(n+1)}{2} - \frac{n+1}{2} \frac{n(n+1)}{2} + n \frac{(n+1)^2}{4} = \sum_{i=1}^n X_i Y_i - \frac{n(n+1)^2}{4} \\ &= \sum_{i=1}^n X_i Y_i - \frac{1}{4} n^3 - \frac{1}{2} n^2 - \frac{1}{4} n \end{aligned}$$

Note that $\sum_{i=1}^n X_i^2 = \sum_{i=1}^n Y_i^2 = \frac{1}{6} n(n+1)(2n+1) = \frac{1}{3} n^3 + \frac{1}{2} n^2 + \frac{1}{6} n$ so the S_{xy} expression can be rewritten as

$$\begin{aligned} &= \frac{1}{12} n(n^2 - 1) + \sum_{i=1}^n X_i Y_i - \frac{1}{2} \sum_{i=1}^n X_i^2 - \frac{1}{2} \sum_{i=1}^n Y_i^2 \\ &= \frac{1}{12} n(n^2 - 1) - \frac{1}{2} \sum_{i=1}^n (X_i - Y_i)^2 \end{aligned}$$

But now

$$r = \frac{12S_{xy}}{n(n^2 - 1)} = \frac{12 \left(\frac{1}{12} n(n^2 - 1) - \frac{1}{2} \sum_{i=1}^n (X_i - Y_i)^2 \right)}{n(n^2 - 1)} = 1 - \frac{6 \sum_{i=1}^n d^2}{n(n^2 - 1)}$$

□

4 Derivations of standard results

Lemma. $\lim_{x \rightarrow \infty} (1 + \frac{a}{x})^{bx} = e^{ab}$ for a, b real

Proof. First note that if $a = 0$ the result is trivially true so we may divide by a . We will show that the limit of the logarithm of $(1 + \frac{a}{x})^{bx}$ exists and equals ab then pass the limit by continuity of e^x . The following limit exists because log is differentiable, and it also equals ab :

$$\begin{aligned} &ab \lim_{\frac{1}{x} \rightarrow 0} \left(\frac{\ln(1 + \frac{a}{x}) - \ln(1)}{\frac{a}{x}} \right) \\ &= ab \lim_{\frac{1}{x} \rightarrow 0} \left(\frac{\ln(1 + \frac{a}{x})}{\frac{a}{x}} \right) = b \lim_{\frac{1}{x} \rightarrow 0} \left(\frac{\ln(1 + \frac{a}{x})}{\frac{1}{x}} \right) = \lim_{x \rightarrow \infty} \left(bx \ln \left(1 + \frac{a}{x} \right) \right) \end{aligned}$$

But this last term is just the limit of the log of the thing that we want to find the limit of. Since ab is the natural logarithm of the limit, the limit is e^{ab} as required.

□

Proposition. The poisson distribution (as we understand it intuitively) satisfies $P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$ where λ is the mean.

Proof. Consider a scenario where the expected number of times for an event to occur in a time interval stays constant but the amount of chances for the event to happen gets larger and the probability of it happening in each time interval gets smaller, and taking a limit. This is what we get. We assume that h is n divided by an integer so that the binomial distribution in question is actually defined.

Imagine making a binomial distribution with arbitrarily short stretches of time but the same mean time per event, in the sense of

$$X \sim B\left(\frac{n}{h}, ph\right)$$

where $h \rightarrow 0$ and $np = \lambda$. We'll consider h to be the reciprocal of an integer just to make this make sense or whatever.

$$P(X = k) = \lim_{h \rightarrow 0} \frac{\left(\frac{n}{h}\right)!}{k! \left(\frac{n}{h} - k\right)!} p^k h^k (1 - ph)^{\frac{n}{h} - k}$$

$$= \lim_{h \rightarrow 0} \frac{\left(\frac{n}{h}\right)^k}{k!} p^k h^k (1 - ph)^{\frac{n}{h} - k}$$

Because as h gets small, $\frac{\left(\frac{n}{h}\right)!}{\left(\frac{n}{h} - k\right)!} = \left(\frac{n}{h}\right) \left(\frac{n}{h} - 1\right) \dots \left(\frac{n}{h} - k + 1\right) \approx \left(\frac{n}{h}\right)^k$ where the ratio between $\frac{\left(\frac{n}{h}\right)!}{\left(\frac{n}{h} - k\right)!}$ and $\left(\frac{n}{h}\right)^k$ approaches 1. Now write the probability as

$$= \lim_{h \rightarrow 0} \frac{n^k}{k!} p^k (1 - ph)^{\frac{n}{h} - k} = \frac{\lambda^k}{k!} \lim_{h \rightarrow 0} (1 - ph)^{\frac{n}{h} - k}$$

Note that $(1 - ph)^{-k}$ goes to 1 as h goes to 0 since p and k are constant, so the probability is

$$\frac{\lambda^k}{k!} \lim_{h \rightarrow 0} (1 - ph)^{\frac{n}{h}}$$

So by the lemma, our final formula is $\frac{e^{-\lambda} \lambda^k}{k!}$. As required. □

In fact, since the poisson probabilities must sum to 1, multiplying by e^λ gives another proof for the exponential function Taylor series.

Corollary. The variance of a Poisson distribution is λ

Proof. In the binomial setting above the variance is $np(1 - ph) \rightarrow np = \lambda$ (By the known variance of the binomial distribution). □

Theorem. The variance of a poisson distribution is λ but this time we won't hand wave the limits.

Proof. Let $X \sim Po(\lambda)$. Then we want to show that $E[X^2] = \lambda + \lambda^2$. We do this as follows:

$$\begin{aligned} E[X^2] &= \sum_{n=0}^{\infty} \frac{e^{-\lambda} \lambda^n}{n!} n^2 = \sum_{n=0}^{\infty} \frac{e^{-\lambda} \lambda^n}{n!} (n(n-1) + n) \\ &= \lambda e^{-\lambda} + \sum_{n=2}^{\infty} \left[\frac{e^{-\lambda} \lambda^n}{(n-2)!} + \frac{e^{-\lambda} \lambda^n}{(n-1)!} \right] = \lambda e^{-\lambda} + \sum_{n=2}^{\infty} \frac{e^{-\lambda} \lambda^n}{(n-2)!} - \lambda e^{-\lambda} + \sum_{n=1}^{\infty} \frac{e^{-\lambda} \lambda^n}{(n-1)!} \\ &= \sum_{n=2}^{\infty} \frac{e^{-\lambda} \lambda^n}{(n-2)!} + \sum_{n=1}^{\infty} \frac{e^{-\lambda} \lambda^n}{(n-1)!} = \sum_{n=0}^{\infty} \frac{e^{-\lambda} (\lambda^{n+2} + \lambda^{n+1})}{n!} = e^{-\lambda} (\lambda^2 + \lambda) \sum_{n=0}^{\infty} \frac{(\lambda^n)}{n!} = \lambda^2 + \lambda \end{aligned}$$

The last equality is by Taylor series for e^λ □

Proposition. A geometric distribution has mean $\frac{1}{p}$ and variance $\frac{1-p}{p^2}$. For negative binomial mean and variance we just multiply by n since the negative binomial is the sum of n geometric distributions.

The convention for the geometric distribution is the same as in level 5.2. That is, a value of n means $n-1$ failures follows by a success.

Proof. Mean of geometric should be intuitive. Although I will give a formal proof, think: If something has a $1/3$ chance of happening, it should take on average 3 tries for it to happen!

Mean of geometric variable x with probability p ($0 < p < 1$) = $\sum_{n=1}^{\infty} n P(x = n) = \sum_{n=1}^{\infty} np(1-p)^{n-1}$ (This is intuitive, it has to happen n times and happen 1 time) = $p \sum_{n=1}^{\infty} n(1-p)^{n-1}$.

This sum is equivalent to adding the rows of the following:

$$\begin{aligned}
& 1 \\
& + (1-p) + (1-p) \\
& + (1-p)^2 + (1-p)^2 + (1-p)^2 \\
& + (1-p)^3 + (1-p)^3 + (1-p)^3 + (1-p)^3 \\
& + \\
& \vdots
\end{aligned}$$

We now consider instead adding the columns rather than the rows.

Technical note: Note that this does converge absolutely, since all terms are non-negative and the sum $\sum_{n=0}^{\infty} n(1-p)^{n-1}$ converges by the ratio test. For details on what this means, see the appendix with technical justification of the generalized binomial theorem in level 4. The reason this matters is that this means we are allowed to change the order in which we sum terms.

Now we add the columns of the image above rather than the rows, giving us that the mean of a geometric random variable with probability p is now equal to (by the geometric series formula)

$$p \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} (1-p)^n (1-p)^m = p \sum_{n=0}^{\infty} \frac{(1-p)^n}{1-(1-p)} = p \sum_{n=0}^{\infty} \frac{(1-p)^n}{p} = \sum_{n=0}^{\infty} (1-p)^n = \frac{1}{1-(1-p)} = \frac{1}{p}$$

as required.

Mean of negative binomial follows from additive property of means, since a negative binomial distribution is the sum of geometric distributions. Variance of negative binomial will follow from additive property of variances when we prove the variance of a geometric distribution, which we will do below:

We know $E[X] = \frac{1}{p}$. Then the variance is:

$$E[X^2] - (E[X])^2 = E[X(X-1)] + E[X] - (E[X])^2 = E[X(X-1)] + \frac{1}{p} - \frac{1}{p^2}$$

Split $E[X^2]$ into $E[X(X-1)] + E[X]$. To determine $E[X(X-1)]$ we will calculate the value of the series $\sum_{k=1}^{\infty} k(k-1)p(1-p)^{k-1}$.

Now we substitute $q = 1-p$.

$$\begin{aligned}
\sum_{k=1}^{\infty} k(k-1)p(1-p)^{k-1} &= p \sum_{k=1}^{\infty} k(k-1)(1-p)^{k-1} \\
&= p \sum_{k=1}^{\infty} k(k-1)q^{k-1} = p \frac{d}{dq} \left(\sum_{k=1}^{\infty} (k-1)q^k \right) \\
&= p \frac{d}{dq} \left(q^2 \sum_{k=1}^{\infty} (k-1)q^{k-2} \right) = p \frac{d}{dq} \left(q^2 \sum_{k=2}^{\infty} (k-1)q^{k-2} \right) \\
&= p \frac{d}{dq} \left(q^2 \frac{d}{dq} \left(\sum_{k=2}^{\infty} q^{k-1} \right) \right) = p \frac{d}{dq} \left(q^2 \frac{d}{dq} \left(\sum_{k=1}^{\infty} q^k \right) \right)
\end{aligned}$$

We could differentiate because we were inside the radius of convergence because the ratio between consecutive terms approaches $1-p$ which is between 0 and 1. Continuing the derivation,

$$= p \frac{d}{dq} \left(q^2 \frac{d}{dq} \left(\frac{1}{1-q} - 1 \right) \right) = p \frac{d}{dq} \left(\frac{q^2}{(1-q)^2} \right)$$

$$= p \left(\frac{-2q}{(q-1)^3} \right) = \left(\frac{-2+2p}{-p^3} \right) = \frac{2(1-p)}{p^2}$$

Therefore $\text{Var}[X] = E[X(X-1)] + \frac{1}{p} - \frac{1}{p^2} = \frac{2(1-p)}{p^2} + \frac{1}{p} - \frac{1}{p^2} = \frac{1-p}{p^2}$

□

5 Probability generating function properties

Proposition. Generating functions multiply when independent variables add

Proof. I will demonstrate why this works with an example. Suppose we have the following random variables x and y:

K	0	1	2	3
P(x=K)	0.1	0.2	0.3	0.4

K	0	1	2
P(y=K)	0.3	0.5	0.2

Then the generating function of x is $0.1 + 0.2t + 0.3t^2 + 0.4t^3$ and the generating function of y is $0.3 + 0.5t + 0.2t^2$. Now consider how we would find the following:

1. The t^4 coefficient of the product of the generating functions
2. $P(x + y = 4)$

For the first one, when we expand it, we will have a t^4 term from the $0.3t^2$ from the first term times the $0.2t^2$ from the second term, and also the $0.4t^3$ term from the first term times the $0.5t$ from the second term. The products that will give a t^4 term are exactly those whose exponents add up to 4.

Now, to find $P(x + y = 4)$, consider all the possible cases: Either $x = 2, y = 2$ or $x = 3, y = 1$. So, since x and y are assumed to be independent, we have that $P(x + y = 4) = P(x = 2)P(y = 2) + P(x = 3)P(y = 1)$. Crucially, multiplying the probabilities that the variable equals k is the same as multiplying the coefficients of t^k by the definition of generating functions, and we do it on exactly those where $x + y$, which corresponds to the sum of the exponents, equals 4.

Hopefully this is convincing enough that the product identity holds.

□

Corollary. Since the negative binomial is the sum of geometric distributions we get the generating function for that for free - it's just the pgf of the geometric distribution to the power of n.

Proposition. If a random variable X has probability generating function $g(t)$ then $aX + b$ has probability generating function $t^b g(t^a)$ provided $a > 0, b \geq 0$ are integers so that $aX + b$ remains non-negative and integer valued.

Proof. The reason this is true is because when we have aX all the possible values for the variables are multiplied by a and these correspond to the exponents in the generating function which therefore must be multiplied by a. Then by adding b we add b to all the possible values for the variables which corresponds to adding b to the exponents in the generating function, thus it is exactly like multiplying each term by t^b .

□

Proposition. $\lim_{z \rightarrow 1^-} p'(z) = E[x]$ when $E[x]$ is finite.

Proof. $p'(z) = \sum r p_r z^{r-1}$

This is increasing and bounded by $E[x]$.

Let $\varepsilon > 0$. Then there is N large enough such that $\sum_{r=1}^n r p_r z^{r-1} \geq E[x] - \varepsilon$ since the infinite sums are the limit of the partial/finite sums and this is how limits are defined.

Now $\lim_{z \rightarrow 1^-} p'(z) \geq \lim_{z \rightarrow 1^-} \sum_{r=1}^n r p_r z^{r-1} \geq E[x] - \varepsilon$

Note that if $|z| < 1$ we are allowed to differentiate term by term due to properties of power series we proved in level 4.

The result follows since ε can be as small as we like.

If $E[x]$ is infinite, then it is the same proof, we say that for each M large enough, $\sum_{r=1}^n r p_r z^{r-1} \geq M$ and we do the same limiting argument.

With the same proof, $\lim_{z \rightarrow 1^-} p''(z) = E[x(x-1)]$ if this limit is finite.

□

6 Unbiased variance estimator, vector approach

Figure 6 shows the geometric intuition for the unbiased variance estimator (ie, $\frac{n}{n-1}$ times the sample variance) using the pythagorean theorem.

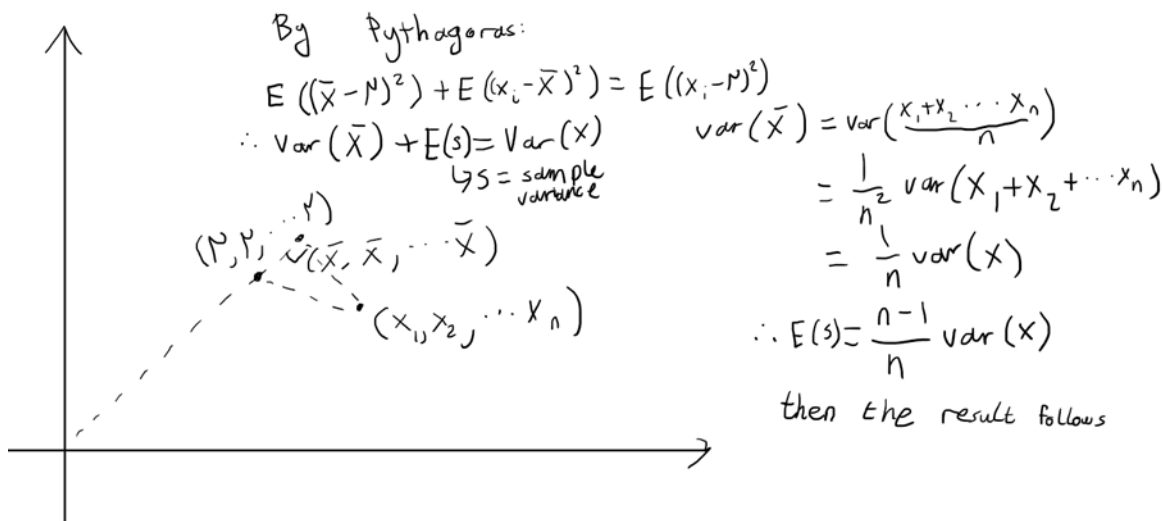


Figure 6

7 Cumulative distribution functions

This chapter is just to note that we have to be careful, because it is not quite true that any non-decreasing function from 0 to 1 that approaches 0 or 1 as x goes to $\pm\infty$ is a cdf.

The reason is that the convention is that the cdf gives $P(X \leq x)$. But then if we have the zero distribution, and we were to say its cdf is 0 at 0 and all negative numbers and 1 at all positive numbers, we would have $P(X \leq 0) = 0$ but $P(X \leq 0.000000000001) = 1$, which is absurd as X cannot take an “infinitesimal” value.

To solve this problem, we impose that a cdf is “right continuous”. What this means is that: For a function to be continuous an equivalent form of the definition is that for every sequence $x_n \rightarrow x$, $f(x_n) \rightarrow f(x)$. Right continuous means this is true if we weaken the assumption to x_n being an arbitrary *decreasing* sequence tending to x .

We know that there are only countably many discontinuities in a cdf by level 6.2. Therefore, the value of the cdf at any

discontinuity point should be the maximum possible.

8 The central limit theorem

8.1 Characteristic functions

Here we will define characteristic functions and prove some basic properties of them (although these basic properties have not such basic proofs). These will play a “central” role in our proof of the CLT. If I have a probability distribution x , then I define the characteristic function of x as $\phi_x(t) = E(e^{itx})$. This is related to the fourier transform of the ditstribution - we will learn about this in later levels.

What we will prove here is that if you know the characteristic function of a distribution you can reverse engineer what the distribution was. To fully understand the proof, you will need to be familiar with level 6.2. The idea will be to show that from the characteristic function there is a formula that lets you reverse engineer the integral from a to b of our probability distribution for any a and b , which determines the entire distribution.

If f is our probability density function (We will generalize this shortly), then

$$\phi_f(t) = E(e^{itx}) = \int_{-\infty}^{\infty} f(y) e^{ity} dy \text{ is the characteristic function.}$$

And define, for arbitrary a and b with $a < b$ and the cdf being continuous at a and b (the last constraint is ok by technical results since the set of discontinuities is countable and thus we can dodge them by moving as small an amount away as we like),

$$I_\varepsilon := \frac{1}{2\pi} \int_{-\infty}^{\infty} \phi_f(t) e^{-\varepsilon t^2} \frac{e^{-iat} - e^{-ibt}}{it} dt$$

The first problem we run into is that not all distributions have a probability density function f , however there is a clean way to solve that problem. I will first mention a possible solution that does not work but I will mention it anyway because it introduces important concepts that will be used in future levels, then I will mention the real solution.

The problem is a distribution may have atoms, which are values a such that $P(x = a) > 0$. Say $P(x = 3) = 0.2$ but otherwise x has a normal probability density function g , or the distribution is discrete. Then we say (in the first case) that $f(x) = g(x) + 0.2\delta(x - 3)$ for some valid density $g(x)$ with total integral 0.8 where $\delta(x)$ is not a function that is defined in the traditional sense, but rather a function that when you integrate it returns 0 if your integration interval does not contain 0 and 1 otherwise. i.e. the following hold:

$$\int_{\mathbb{R}} \delta(x) h(x) dx = h(0) \text{ where this notation means integrating over all real numbers, i.e. minus infinity to infinity. Also,}$$

$$\int_a^b \delta(x) dx = \begin{cases} 1 & 0 \in [a, b] \\ 0 & \text{otherwise} \end{cases}$$

This is called the dirac delta function. You can think of it as a spike at 0 and 0 otherwise, or as the limit of normals with mean 0 and tinier variances approaching 0.

However, this is not enough, as not all distributions are part discrete part continuous. The solution is to integrate with respect to the probability law.

Why this is necessary: We could say we just want to prove the CLT for part discrete part continuous distributions and then stick to $f(x)$ notation. For 4 months, that is what this document did. However, this is wrong because it implicitly assumes the limiting distribution is of this type.

Example:

$$\int_a^b d\mu = P(a < x < b)$$

$$\int_{-\infty}^{\infty} x d\mu = E(x)$$

We can think of $d\mu$ as $f(x)dx$ where $f(x)$ is a density, it's just that a density may not always exist, but this new length system can be thought of (when the distribution is continuous) as rescaling the number line by the density so that it ends

up having length 1 and our distribution is uniform on it.

$$\int_{-\infty}^{\infty} x^2 d\mu - E(x)^2 = \text{Var}(x)$$

$$\phi_y(t) = E(e^{ity}) = \int_{-\infty}^{\infty} e^{ity} d\mu$$

We will now use this notation.

Lemma. Characteristic functions are continuous, and in fact they are uniformly continuous.

Proof. In this proof I will use the inequality $|e^{ix} - e^{iy}| < |x - y|$ if x and y are real numbers not equal to each other. The reason this inequality is true can be shown geometrically (Figure 7):

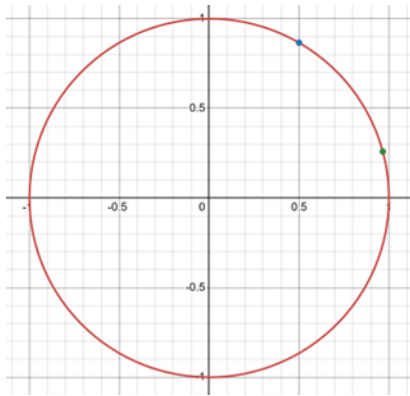


Figure 7

The image above shows a unit circle on the complex plane with 2 points marked to demonstrate the “shortest distance” principle

Here, the green and blue points represent values of e^{ix} for certain values of x . The distance that they are apart along the unit circle is exactly the difference between these values of x , but the distance between these points is less than this difference, because you could get further by walking in a straight line, as that is more efficient. That is why this inequality holds.

Given $\eta > 0$ pick M such that $P(|X| > M) < \frac{\eta}{4}$. Then

$$|\phi_X(s) - \phi_X(t)| \leq E[|e^{isX} - e^{itX}|] \leq M|s - t| + 2P[|X| > M]$$

Taking $|s - t| < \frac{\eta}{2M}$ makes the result less than η and since this did not depend on the choice of s, t the continuity is uniform.

□

Now, we come to the hardest theorem yet on this website, a title which will be dethroned in the next section of this very same document.

Theorem. The characteristic function determines the distribution.

Proof. The idea is to try to find a way to determine $P(a < X < b)$, and the dominated convergence theorem is a tool that lets us do this using a limit.

Recall the definition of I_ϵ from earlier. Now we can say that

$$I_\epsilon = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} e^{ity} d\mu \right] e^{-\epsilon t^2} \frac{e^{-iat} - e^{-ibt}}{it} dt$$

Now we will show that

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \left| e^{ity} e^{-\varepsilon t^2} \frac{e^{-iat} - e^{-ibt}}{it} \right| d\mu dt$$

is finite so that we can swap the integrals around. Note that when $t = 0$, L'hopital's rule implies that we should set by convention $\frac{e^{-iat} - e^{-ibt}}{it} = b - a$ to preserve continuity so we don't have a point of undefinedness. Note that for each fixed t ,

$$\begin{aligned} \int_{-\infty}^{\infty} \left| e^{ity} e^{-\varepsilon t^2} \frac{e^{-iat} - e^{-ibt}}{it} \right| d\mu &= \int_{-\infty}^{\infty} |e^{ity}| |e^{-\varepsilon t^2}| \left| \frac{e^{-iat} - e^{-ibt}}{it} \right| d\mu \\ &= |e^{-\varepsilon t^2}| \left| \frac{e^{-iat} - e^{-ibt}}{it} \right| \int_{-\infty}^{\infty} |e^{ity}| d\mu = |e^{-\varepsilon t^2}| \left| \frac{e^{-iat} - e^{-ibt}}{it} \right| \end{aligned}$$

so it remains to show that $\int_{-\infty}^{\infty} |e^{-\varepsilon t^2}| \left| \frac{e^{-iat} - e^{-ibt}}{it} \right| dt$ is finite (as the factor of 2π does not change finiteness), which simplifies our problem massively.

To do this, I will put some bounds on the term $\left| \frac{e^{-iat} - e^{-ibt}}{it} \right|$. This term is equal to $\frac{|e^{-iat} - e^{-ibt}|}{|t|}$.

So, if $|t| < 1$, our expression is less than $\frac{|at - bt|}{|t|}$ which is just $|a - b|$. If $|t| \geq 1$, then the numerator is the absolute value of the difference between two points on the unit circle, which cannot be more than the diameter of the unit circle which is 2, so it is at most $\frac{2}{|t|}$. So, the integral $\int_{-\infty}^{\infty} |e^{-\varepsilon t^2}| \left| \frac{e^{-iat} - e^{-ibt}}{it} \right| dt$ is at most this:

$$\int_{|t| \leq 1} |b - a| e^{-\varepsilon t^2} dt + \int_{|t| > 1} \frac{2e^{-\varepsilon t^2}}{|t|} dt$$

The first term is at most $2|b - a|$. The second term will just get larger if you remove the denominator, and then it turns into an integral which we have already proven has a finite value, which we could give in terms of the square root of pi.

Now we get this after swapping the integrals and combining some exponents:

$$I_\varepsilon = \int_{-\infty}^{\infty} K_\varepsilon(y) d\mu \text{ where } K_\varepsilon(y) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-\varepsilon t^2} \frac{e^{it(y-a)} - e^{it(y-b)}}{it} dt$$

Definition: $\text{erf}(x) := \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$. This function goes to 1 as x goes to infinity and -1 as x goes to negative infinity. It is essentially a cdf of the normal rescaled to go from -1 to 1.

I will now define another new function as follows:

$$H_\varepsilon(u) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-\varepsilon t^2} \frac{e^{itu} - 1}{it} dt$$

Note that $H_\varepsilon(u) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-\varepsilon t^2} \int_0^u e^{its} ds dt$.

We want to show that $\int_{-\infty}^{\infty} \left| \int_0^u |e^{-\varepsilon t^2} e^{its}| ds \right| dt$ is finite so that we can swap the integrals around. The second set of absolute values is necessary because u could be negative in which case all the signs would be flipped.

$$\int_{-\infty}^{\infty} \left| \int_0^u |e^{-\varepsilon t^2} e^{its}| ds \right| dt = \int_{-\infty}^{\infty} \left| \int_0^u |e^{-\varepsilon t^2}| |e^{its}| ds \right| dt = \int_{-\infty}^{\infty} |e^{-\varepsilon t^2}| \left| \int_0^u |e^{its}| ds \right| dt = \int_{-\infty}^{\infty} e^{-\varepsilon t^2} \left| \int_0^u 1 ds \right| dt = \int_{-\infty}^{\infty} |u| e^{-\varepsilon t^2} dt.$$

If we use the substitution $v = t\sqrt{\varepsilon}$ it will follow that this integral is $|u| \sqrt{\frac{\pi}{\varepsilon}}$ which is finite. Therefore we get that $H_\varepsilon(u)$ is equal to this:

$$\frac{1}{2\pi} \int_0^u \left(\int_{-\infty}^{\infty} e^{-\varepsilon t^2} e^{ist} dt \right) ds$$

We now complete the square as follows:

$$-\varepsilon t^2 + ist = -\varepsilon \left(t - \frac{is}{2\varepsilon} \right)^2 - \frac{s^2}{4\varepsilon}$$

To get

$$\int_{-\infty}^{\infty} e^{-\varepsilon t^2} e^{ist} dt = e^{-\frac{s^2}{4\varepsilon}} \int_{-\infty}^{\infty} e^{-\varepsilon(t - \frac{is}{2\varepsilon})^2} dt = \sqrt{\frac{\pi}{\varepsilon}} e^{-\frac{s^2}{4\varepsilon}}$$

The reason we can shift by an imaginary constant step and still get the same result is because our function has a global single valued antiderivative defined by the Taylor series which converges everywhere, so you can integrate along contours like these and get the same result

Figure 8 shows a contour that is the reals shifted by an imaginary constant for a large interval



Figure 8

As we make the red part longer, we know from considering what function we are actually integrating that the area of the tails and along the vertical parts will go to 0, so the limit of these integrals which are all the same as the integral along the real line will be the same as the integral along the real line and be equal to the integral with the imaginary shift.

Therefore, $H_\varepsilon(u)$ is equal to $\frac{1}{2\pi} \sqrt{\frac{\pi}{\varepsilon}} \int_0^u e^{-\frac{s^2}{4\varepsilon}} ds$, which can be shown by the substitution $v = \frac{s}{2\sqrt{\varepsilon}}$ to be equal to $\frac{1}{2} \operatorname{erf}(\frac{u}{2\sqrt{\varepsilon}})$.

Now we have the following identity from the definitions:

$$K_\varepsilon(y) = H_\varepsilon(y-a) - H_\varepsilon(y-b) = \frac{1}{2} \left(\operatorname{erf}\left(\frac{y-a}{2\sqrt{\varepsilon}}\right) - \operatorname{erf}\left(\frac{y-b}{2\sqrt{\varepsilon}}\right) \right)$$

Now consider what the K function approaches as ε approaches 0: If $y < a$, then y also is $< b$ since we defined $a < b$. Therefore, both terms will go to -1 as the inputs to the erf functions will go to -infinity, so the whole thing will go to 0. Same if $y > b$. But, if $a < y < b$, then the first term's input will go to infinity so the first term will go to 1, and the second term's input will go to -infinity so the whole thing will go to $\frac{1}{2}(1 - (-1))$ which is 1. Therefore, as ε goes to 0, K approaches the function that is 1 when y is between a and b and 0 otherwise, and it approaches $\frac{1}{2}$ at exactly a and b , but this doesn't really matter. What does matter is that K is always between -1 and 1.

Recall, we had this:

$$I_\varepsilon = \int_{-\infty}^{\infty} K_\varepsilon(y) d\mu \text{ where } K_\varepsilon(y) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-\varepsilon t^2} \frac{e^{it(y-a)} - e^{it(y-b)}}{it} dt$$

We know that since K is between 0 and 1, define $f_n(y) := K_{\frac{1}{n}}(y)$, then $f_n(y)$ is bounded above in absolute value by 1, so dominated convergence applies (this is a genuine dominating function with a finite integral as we are integrating with respect to a finite measure). Since $f_n(y)$ converges pointwise to 1 when $a < y < b$ and 0 otherwise (and what happens at exactly a and b does not affect the value of the integrals since we assume a, b to be continuity points), we know by dominated convergence on f_n that as ε goes to 0, I_ε goes to $\int_a^b d\mu$. But I_ε was defined only in terms of the characteristic function and we just determined $\int_a^b d\mu$ where a and b were chosen arbitrarily.

□

Lemma. Convergence in distribution implies pointwise convergence of characteristic function

Proof. Suppose we have a sequence of probability distributions X_n converging in distribution to a probability distribution X . For any $\varepsilon > 0$, we can pick a point on the cdf of X with a value strictly less than $\frac{\varepsilon}{2}$, and another point with a value strictly greater than $1 - \frac{\varepsilon}{2}$. This is always possible since in the extremes, the cdf approaches 0 and 1, so we can find points as close as we want to 0 and 1. Let M be at least the maximum absolute value of these points we've picked and ensure we pick M such that at M and $-M$, the cdf of X and each X_n is continuous and does not jump, possible as there are at

most countably many total discontinuities. Then $P(|X| > M) < \varepsilon$. But since X_n converges in distribution to X , the cdfs converge pointwise, so at $-M$ and M where the cdf of X is within $\frac{\varepsilon}{2}$ of 0 and 1 respectively, by definition of convergence we can find an N such that for all X_n with $n > N$, the cdf of X_n at M and $-M$ get as close as we want to the cdf of X at M and $-M$. Specifically, make it so close that it is still within $\frac{\varepsilon}{2}$ of 0 and 1 respectively. Then we have shown that for any $\varepsilon > 0$ there exists M and N such that if $n > N$ then $P(X_n \in (-M, M]) < \varepsilon$.

Now for any $\varepsilon > 0$ pick M and N_1 such that if $n > N_1$ we have that $P(X_n \in (-M, M]) < \frac{\varepsilon}{8}$ and $P(X \in (-M, M]) < \frac{\varepsilon}{8}$. Now let 1_S be the indicator function of the set S , i.e. the function that returns 1 for points inside the set S and 0 everywhere else. Then

$$E(e^{itX_n}) - E(e^{itX}) = E(e^{itX_n} 1_{X_n \in (-M, M]} + e^{itX_n} 1_{X_n \notin (-M, M]}) - E(e^{itX} 1_{X \in (-M, M]} + e^{itX} 1_{X \notin (-M, M]})$$

We simply have that the expectation overall is the expectation when inside a set plus the expectation when outside that set, this is obvious. We split this even further to get the following:

$$= E(e^{itX_n} 1_{X_n \in (-M, M]}) + E(e^{itX_n} 1_{X_n \notin (-M, M]}) - E(e^{itX} 1_{X \in (-M, M]}) - E(e^{itX} 1_{X \notin (-M, M]})$$

Now, recall that we have the triangle inequality for both real numbers and integrals. Since expectations are actually defined in terms of integrals, we do have the inequality $|E(x)| \leq E(|x|)$ directly from the integral inequality. So, we have:

$$\begin{aligned} |E(e^{itX_n}) - E(e^{itX})| &\leq |E(e^{itX_n} 1_{X_n \in (-M, M]}) - E(e^{itX} 1_{X \in (-M, M]})| + |E(e^{itX_n} 1_{X_n \notin (-M, M]})| + |E(e^{itX} 1_{X \notin (-M, M]})| \\ &\leq |E(e^{itX_n} 1_{X_n \in (-M, M]}) - E(e^{itX} 1_{X \in (-M, M]})| + E(|e^{itX_n}| 1_{X_n \notin (-M, M]}) + E(|e^{itX}| 1_{X \notin (-M, M]}) \end{aligned}$$

By the triangle inequality for expectations. However, $|e^{itX_n}| = |e^{itX}| = 1$ so since this is only when $|X_n| > M$, we have that

$$|E(e^{itX_n}) - E(e^{itX})| \leq |E(e^{itX_n} 1_{X_n \in (-M, M]}) - E(e^{itX} 1_{X \in (-M, M]})| + \frac{\varepsilon}{4}$$

Which helps a lot, since the goal is to make this term less than a full epsilon so we have that the characteristic functions get as close together as we want so we have the desired result.

Now, e^{itx} on $[-M, M]$ is a continuous function on a closed bounded interval so it is uniformly continuous and thus also uniformly continuous on $(-M, M]$. Therefore we can find a δ small enough that on any interval of length at most δ , e^{itx} is within a ball in the complex plane of diameter $\frac{\varepsilon}{4}$, which we can do by considering real and imaginary parts separately and making sure they are in an interval of length $\frac{\varepsilon}{4\sqrt{2}}$, ensuring the whole thing is in a square within the circle that we need. Since this δ works everywhere by uniform continuity, we can find a finite partition $-M = a_0 < a_1 < \dots < a_m = M$ where the distance between each a_i is at most δ . We take care to make sure that each a is a point where the cdfs of all X_n, X are continuous (since there are at most countably many total discontinuities), and this will have the property that if $g(x) := e^{itx}$, then for any x and y between a_{j-1} and a_j we have that $|g(x) - g(y)| < \frac{\varepsilon}{4}$. In particular, for any ξ_j with $a_{j-1} < \xi_j \leq a_j$, which I will now pick arbitrarily for each interval j , we have that the difference between $c_j := g(\xi_j)$ and $g(\text{anything else in that interval})$ is bounded above in absolute value by $\frac{\varepsilon}{4}$. I will now define the simple function $s(x) := \sum_{j=1}^m c_j 1_{(a_{j-1}, a_j]}$, then since $|c_j|$ is 1 since it is equal to $e^{i \cdot \text{stuff}}$, we have that it is always the case that $|g(x) - s(x)| < \frac{\varepsilon}{4}$. Now lets work on the $|E(e^{itX_n} 1_{X_n \in (-M, M]}) - E(e^{itX} 1_{X \in (-M, M]})|$ term which we hope to bound by $\frac{3\varepsilon}{4}$ in order to be done: This term is equal to $|\int_{(-M, M]} g(x) d\mu_n - \int_{(-M, M]} g(x) d\mu|$ because of the definition of expected value in terms of integrals and the definition of $g(x)$ and the fact that the indicator function makes this be on the interval $[-M, M]$. I can write the term as

$$\left| \int_{(-M, M]} (g - s)(x) d\mu_n - \int_{(-M, M]} (g - s)(x) d\mu + \int_{(-M, M]} s(x) d\mu_n - \int_{(-M, M]} s(x) d\mu \right|$$

Then both versions of the triangle inequality mean that this is

$$\leq \int_{(-M, M]} |(g - s)(x)| d\mu_n + \int_{(-M, M]} |(g - s)(x)| d\mu + \left| \int_{(-M, M]} s(x) d\mu_n - \int_{(-M, M]} s(x) d\mu \right|$$

Since $|g - s|$ is positive and real so we can pull it out of the absolute value. Now using the bounds we got earlier we can simplify this even further:

$$\begin{aligned} &\leq \int_{(-M, M]} \frac{\varepsilon}{4} d\mu_n + \int_{(-M, M]} \frac{\varepsilon}{4} d\mu + \left| \int_{(-M, M]} s(x) d\mu_n - \int_{(-M, M]} s(x) d\mu \right| \\ &\leq \frac{\varepsilon}{2} + \left| \int_{(-M, M]} s(x) d\mu_n - \int_{(-M, M]} s(x) d\mu \right| \end{aligned}$$

Since those integrals are with respect to a probability measure and thus each bounded by $\frac{\varepsilon}{4}$. So we just have to make sure that

$$\left| \int_{(-M, M]} ss(x) d\mu_n - \int_{(-M, M]} ss(x) d\mu \right| \leq \frac{\varepsilon}{4}$$

then we will be done, since we will have that the characteristic function converges for every fixed t and therefore converges pointwise.

To do this, we note that since s is equal to c_j on each interval (a_{j-1}, a_j) , we have that the term in question can be written as

$$\left| \sum_{j=1}^m \int_{a_{j-1}}^{a_j} c_j d\mu_n - \sum_{j=1}^m \int_{a_{j-1}}^{a_j} c_j d\mu \right|$$

Call the corresponding cdfs F_n and F respectively, so the term becomes

$$\begin{aligned} &\left| \sum_{j=1}^m c_j [F_n(a_j) - F_n(a_{j-1})] - \sum_{j=1}^m c_j [F(a_j) - F(a_{j-1})] \right| \\ &= \left| \sum_{j=1}^m c_j [(F_n(a_j) - F_n(a_{j-1})) - (F(a_j) - F(a_{j-1}))] \right| \\ &\leq \sum_{j=1}^m |c_j| |(F_n(a_j) - F_n(a_{j-1})) - (F(a_j) - F(a_{j-1}))| \end{aligned}$$

By the triangle inequality

$$= \sum_{j=1}^m |[(F_n(a_j) - F_n(a_{j-1})) - (F(a_j) - F(a_{j-1}))]|$$

Since $|c_j| = 1$

$$\leq \sum_{j=1}^m [|F_n(a_j) - F(a_j)| + |F_n(a_{j-1}) - F(a_{j-1})|]$$

By the triangle inequality.

Now, since each a_j is a point where F is continuous, it means F_n converges to F there. So if n is large enough $F_n(a_j)$ will be within $\frac{\varepsilon}{8m}$ of $F(a_j)$ if we pick $n > N_j$. Since there are finitely many (m , the number of intervals in our partition) N_j 's, simply pick the largest one, then we have that n is large enough so that our sum is

$$\leq \sum_{j=1}^m \left[\frac{\varepsilon}{8m} + \frac{\varepsilon}{8m} \right] = \sum_{j=1}^m \left[\frac{\varepsilon}{4m} \right] = \frac{\varepsilon}{4}$$

□

Theorem. (Levy's theorem) If the characteristic function of a sequence of probability distributions converges to a function which is the characteristic function of some distribution, then the distributions converge to the probability distribution with the limit characteristic function.

We use the following lemma.

Lemma. (Helly's selection theorem) Suppose we have an open interval in the real line or the real line itself, which we will call I . Let (f_n) be a sequence of increasing functions $I \rightarrow \mathbb{R}$ such that there exists a and b such that for all n , $a \leq f_n \leq b$. Then the sequence (f_n) has a subsequence of functions $I \rightarrow \mathbb{R}$ which converges at each point to some function $f : I \rightarrow \mathbb{R}$.

Proof. We note that an increasing function has countably many discontinuity points and countably many rational points. Enumerate all the rational points and points where some f_n is discontinuous by $a_1, a_2, \dots$ then by the Bolzano-Weierstrass theorem there is a subsequence of functions that converges at a_1 , we'll call this $f_{1,n}$. Of this subsequence, there is a further subsequence $f_{2,n}$ that also converges at a_2 . We can keep making these nested subsequences, then the sequence $f_{n,n}$ converges at every rational point. Call this subsequence g_k , then it converges at all rational points and discontinuity points of some f_n .

That's basically the idea. We now want to extend convergence on this countable set to convergence in I . Call the countable set that g_k converges in A . Since limits preserve non-strict inequalities, limits of increasing functions are increasing so the limit of g_k is an increasing function and thus has at most countably many discontinuities. Define

$$h_k(x) := \sup_{a \leq x, a \in A} g_k(a)$$

Then $h_k(x) = g_k(x)$ whenever $x \in A$ and h_k is an increasing function. Define

$$h(x) = \limsup_{k \rightarrow \infty} h_k(x)$$

Since limits and suprema of increasing functions are increasing, h is increasing and in fact when x is in A it is just the limit of g_k . Therefore $g_k \rightarrow h$ in A . We do all this because it is useful to know that in A the limiting function of g is actually equal to $\limsup_{k \rightarrow \infty} g_k(x)$

We want to show that g_k converges at all continuity points of h .

Let x be a continuity point of h and (by openness) let $q < x < r$ and let $q, r \in A$. Then by increasingness of everything we have

$$\begin{aligned} g_k(q) - h(r) &\leq g_k(x) - h(x) \leq g_k(r) - h(q) \\ \limsup_{k \rightarrow \infty} (g_k(x) - h(x)) &\leq \limsup_{k \rightarrow \infty} (g_k(r) - h(q)) \leq h(r) - h(q) \\ h(q) - h(r) &= \liminf_{k \rightarrow \infty} (g_k(q) - h(r)) \leq \liminf_{k \rightarrow \infty} (g_k(x) - h(x)) \\ h(q) - h(r) &\leq \liminf_{k \rightarrow \infty} (g_k(x) - h(x)) \leq \limsup_{k \rightarrow \infty} (g_k(x) - h(x)) \leq h(r) - h(q) \end{aligned}$$

Since h is continuous at x and there are points in A in any arbitrarily small open interval around x , passing the limit as $q \rightarrow x$ and $r \rightarrow x$ we have $h(q), h(r) \rightarrow h(x)$. Limits preserve non-strict inequalities, and in the limit the first and last thing in our chain of inequalities is 0. Therefore, both $\liminf_{k \rightarrow \infty} (g_k(x) - h(x)) = 0$ and $\limsup_{k \rightarrow \infty} (g_k(x) - h(x)) = 0$. This establishes $g_k \rightarrow h$ at all continuity points of h .

Since there are countably many non-converging points, we use another level-6.4-style argument to find a further subsequence of g_k that converges at all of those discontinuities.

□

Now we will prove Levy's theorem.

Definition. A sequence of random variables X_n is said to be tight if for all $\varepsilon > 0$ there exists an $M > 0$ such that $P(|X_n| > M) < \varepsilon$. Clearly a finite set of random variables is tight, but for infinite sequences justification is required.

Now here's the proof of Levy's theorem.

Proof. We will define yet another function. Let $L_\varepsilon(t) := \frac{e^{-\varepsilon t^2}}{2\pi} \frac{e^{-iat} - e^{-ibt}}{it}$, then $I_\varepsilon = \int_{-\infty}^{\infty} \phi(t) L_\varepsilon(t) dt$. L_ε is integrable with the same proof as earlier and bounded by $\frac{b-a}{2\pi}$, so if our sequence of characteristic functions $\phi_n(t)$ converges pointwise to $\phi(t)$ then $\phi_n(t) L_\varepsilon(t)$ converges to $\phi(t) L_\varepsilon(t)$ as n goes to infinity. Also, $|\phi_n(t)| \leq 1$ from earlier, so $|\phi_n(t) L_\varepsilon(t)| < |L_\varepsilon(t)|$. Therefore all the hypotheses for the dominated convergence theorem apply, so the limit of $\int_{-\infty}^{\infty} \phi_n(t) L_\varepsilon(t) dt$ is indeed

I_ε . For a year, I thought this was where we were done. However, unfortunately this is not quite the case, and I believe it is worth explaining exactly why because as a mathematician one needs to avoid subtle errors like this.

Set the integral of $\phi_n(t) L_\varepsilon(t)$ to be called $I_{n,\varepsilon}$.

The issue with the above proof is that we only know $I_\varepsilon \rightarrow P[a < x < b]$ and that for each fixed ε we have

$$\lim_{n \rightarrow \infty} I_{n,\varepsilon} = I_\varepsilon$$

and we also have that

$$\lim_{\varepsilon \rightarrow 0} \lim_{n \rightarrow \infty} I_{n,\varepsilon} = \lim_{\varepsilon \rightarrow 0} I_\varepsilon$$

However we actually want to show that

$$\lim_{n \rightarrow \infty} \lim_{\varepsilon \rightarrow 0} I_{n,\varepsilon} = \lim_{\varepsilon \rightarrow 0} I_\varepsilon$$

To actually prove the named theorem, a different approach is useful. We bypass swapping the limits above by using Helly's selection theorem, but I chose to keep that in since it's very subtle and therefore worth seeing from a learning standpoint.

Let F_n be a sequence of CDFs of random variables X_n with characteristic functions $\phi_n(t)$ and corresponding probability measures notated as $d\mu_n$ such that for all t , $\phi_n(t) \rightarrow \phi(t)$ and $\phi(t)$ is a characteristic function of some distribution and thus continuous.

Let $u > 0$. Then we have

$$\frac{1}{2u} \int_{-u}^u (1 - \phi_n(t)) dt = \frac{1}{2u} \int_{-u}^u \left(\int_{-\infty}^{\infty} 1 - e^{itx} d\mu_n \right) dt$$

We have absolute convergence because the integral of the absolute value is finite, as it looks like we have an infinite domain but we are integrating something bounded against a probability measure so the result is actually finite. Therefore, with care, we can swap the integrals around and say

$$= \int_{-\infty}^{\infty} \left(\frac{1}{2u} \int_{-u}^u (1 - e^{itx}) dt \right) d\mu_n = \int_{-\infty}^{\infty} \left(1 - \frac{e^{iux} - e^{-iux}}{2ux} \right) d\mu_n = \int_{-\infty}^{\infty} \left(1 - \frac{\sin(ux)}{ux} \right) d\mu_n$$

If $|ux| \geq 2$ then we have $\left| \frac{\sin(ux)}{ux} \right| \leq \frac{1}{2}$. Therefore, the term $1 - \frac{\sin(ux)}{ux}$ is $> \frac{1}{2}$ when $|x| \geq \frac{2}{u}$ (since if x is exactly equal to that then $\sin(2) < 1$ lets the inequality be strict). Therefore, we have

$$\int_{-\infty}^{\infty} \left(1 - \frac{\sin(ux)}{ux} \right) d\mu_n \geq \int_{|X_n| \geq \frac{2}{u}} \left(\frac{1}{2} \right) d\mu_n = \frac{1}{2} P\left(|X_n| \geq \frac{2}{u}\right)$$

All this to say we have

$$P\left(|X_n| \geq \frac{2}{u}\right) \leq \frac{1}{u} \int_{-u}^u (1 - \phi_n(t)) dt$$

Fix $\varepsilon > 0$. Because $\phi_n \rightarrow \phi$ pointwise and $\phi(0) = 1$ and ϕ is continuous there exists a sufficiently small $u > 0$ such that $\frac{1}{u} \int_{-u}^u (1 - \phi(t)) dt < \frac{\varepsilon}{2}$. Since $|1 - \phi_n(t)| \leq 2$, the dominated convergence theorem implies that

$$\frac{1}{u} \int_{-u}^u (1 - \phi_n(t)) dt \rightarrow \frac{1}{u} \int_{-u}^u (1 - \phi(t)) dt < \frac{\varepsilon}{2}$$

Thus if n is sufficiently large we conclude that $P\left(|X_n| \geq \frac{2}{u}\right) < \varepsilon$. This establishes that X_n is tight since the max over all n of M such that $P(|X_i| > M) < \varepsilon$ is the max between $\frac{2}{u}$ and finitely many other constants.

The reason we established tightness is so that when we later apply Helly's selection theorem, our limit function is actually a probability distribution that approaches 0 and 1 as x goes to $\pm\infty$.

We have that there is a subsequence F_{n_k} converging to a genuine cumulative distribution function F at each continuity point.

Since this implies the characteristic functions converge, the characteristic functions of F_{n_k} must converge to the characteristic function of F , which therefore must be ϕ , so F is the cdf of the distribution with characteristic function ϕ because characteristic functions determine distribution. We know that every arbitrary subsequence F_{n_j} has a further subsequence

$F_{n_{j_l}}$ converging to the cdf of the distribution with characteristic function ϕ , and if convergence in distribution did not occur then it would be possible to exhibit a subsequence with no such subsequence which is a contradiction.

□

Note: This means the distribution converges if the cf converges, in the sense that the weight of the distribution on any interval converges, but the probability density function need not converge pointwise. Convergence in distribution actually means the cumulative distribution function converges pointwise at all points where it is continuous, which this satisfies since we showed that all integrals of the pdf converge so the cdf converges.

8.2 Proof of CLT

We will prove this for distributions that have a probability density function that have finite variance. The cauchy distribution is an example of something that does not work due to not having finite variance – In fact for the cauchy distribution the CLT is not even true.

Setup: Let $X_1, X_2, \dots$ be identically distributed and independent where each has mean 0 and variance 1. We can start with any sequence of iid random variables and scale/shift them appropriately for this to be the case.

Define the normalized sum $S_n = \frac{X_1 + \dots + X_n}{\sqrt{n}}$.

Most of the difficulty of the proof was in the properties of characteristic functions. It turns out that those were the “technical details” and the main argument is quite short.

Theorem. (The central limit theorem) S_n converges in distribution (in the sense of convergence of the probability of it falling within any interval with end points being continuity points of the cumulative distribution function) to $N(0, 1)$.

Proof. With our theorems in hand, we just need to show that $\phi_{S_n}(t) \rightarrow \phi_{N(0,1)}(t)$ as $n \rightarrow \infty$.

The cf of a standard normal can be calculated as follows.

$$\begin{aligned} \int_{-\infty}^{\infty} \frac{e^{-\frac{y^2}{2}}}{\sqrt{2\pi}} e^{ity} dy &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{y^2}{2}} e^{ity} dy = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(y^2 - 2ity)} dy = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}((y-it)^2 - t^2)} dy \\ &= \frac{1}{\sqrt{2\pi}} e^{-t^2/2} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(y-it)^2} dy = e^{-t^2/2} \end{aligned}$$

Where we have used the same contour integration idea as in the characteristic function proof.

Now we just need to show that $\phi_{S_n}(t) \rightarrow e^{-\frac{t^2}{2}}$.

S_n has mean 0 and variance 1 so it is standardized, and the goal is to prove that as n gets large, S_n is well approximated by a standard normal, as this is what the central limit theorem says. The characteristic function of S_n is

$$E\left(e^{it\left(\frac{X_1}{\sqrt{n}}\right) + it\left(\frac{X_2}{\sqrt{n}}\right) + \dots + it\left(\frac{X_n}{\sqrt{n}}\right)}\right) = E\left(e^{it\left(\frac{X_1}{\sqrt{n}}\right)}\right) E\left(e^{it\left(\frac{X_2}{\sqrt{n}}\right)}\right) \dots E\left(e^{it\left(\frac{X_n}{\sqrt{n}}\right)}\right)$$

by properties of exponents and the fact that with independent things we can split expectation and product. Therefore, $\phi_{S_n}(t) = \left(\phi_x\left(\frac{t}{\sqrt{n}}\right)\right)^n$. We will prove that $\phi_x(u) = 1 - \frac{u^2}{2} + o(u^2)$. We will then be able to put $u_n = \frac{t}{\sqrt{n}}$ to get $\phi_x(u_n) = 1 - \frac{t^2}{2n} + o\left(\frac{1}{n}\right)$.

Note that here, some proofs expand e^{itx} as a Taylor series, but this is not justified since after this a swap of expectation and infinite summation that is not justified normally takes place. Below is the rigorous way to do this.

Define for real y

$$R(y) := e^{iy} - 1 - iy + \frac{y^2}{2}$$

$$\int_0^1 (1-s)e^{isy} ds = \left[(1-s) \frac{e^{isy}}{iy} \right]_0^1 + \int_0^1 \frac{e^{isy}}{iy} ds = -\frac{1}{iy} + \left[\frac{e^{isy}}{(iy)^2} \right]_0^1 = -\frac{1}{iy} + \frac{e^{iy}}{(iy)^2} - \frac{1}{(iy)^2}$$

Where I have used integration by parts with $u = 1-s$ and $v = \frac{e^{isy}}{iy}$

Therefore $e^{iy} - 1 - iy = (iy)^2 \int_0^1 (1-s)e^{isy} ds$

So

$$\begin{aligned} |e^{iy} - 1 - iy| &= |(iy)^2| \left| \int_0^1 (1-s)e^{isy} ds \right| \leq y^2 \int_0^1 |(1-s)e^{isy}| ds \\ &= y^2 \int_0^1 |1-s| |e^{isy}| ds = y^2 \int_0^1 |1-s| ds = \frac{y^2}{2}. \end{aligned}$$

Therefore,

$$|R(y)| \leq |e^{iy} - 1 - iy| + \frac{y^2}{2} \leq y^2$$

Now notice that

$$\begin{aligned} (iy)^2 \int_0^1 (1-s)(e^{isy} - 1) ds &= e^{iy} - 1 - iy - (iy)^2 \int_0^1 (1-s) ds \\ &= e^{iy} - 1 - iy + y^2 \int_0^1 (1-s) ds = e^{iy} - 1 - iy + \frac{y^2}{2} = R(y) \end{aligned}$$

So,

$$|R(y)| = \left| (iy)^2 \int_0^1 (1-s)(e^{isy} - 1) ds \right| \leq y^2 \int_0^1 (1-s)s |y| ds = \frac{|y|^3}{6}$$

Now define h for real numbers as follows:

$$h(y) = \begin{cases} \frac{R(y)}{y^2} & y \neq 0 \\ 0 & y = 0 \end{cases}$$

By our two bounds on $R(y)$, h is always bounded by 1, and since it is bounded by $y/6$ it goes to 0 as y goes to 0.

Now, using $E(X) = 0$ and therefore that $E(x^2) = E(x^2) - E(x)^2 = 1$, we have that

$$\phi(u) = E[e^{iuX}] = 1 + iuE[X] - \frac{u^2}{2}E[X^2] + E[R(uX)] = 1 - \frac{u^2}{2} + E[R(uX)]$$

Now, by definition of h ,

$$\frac{\phi(u) - 1 + \frac{u^2}{2}}{u^2} = E \left[\frac{R(uX)}{u^2} \right] = E[h(uX)X^2]$$

Therefore if this goes to 0 as u goes to 0, we have $o(u^2)$ so we will be done with finding the cf of x .

We want to prove $\lim_{u \rightarrow 0} \frac{1}{u^2} E[R(ux)] = 0$ where $R(y) = e^{iy} - 1 - iy + \frac{y^2}{2}$ and $E[R(ux)] = \int_0^1 R(ux) d\mu$

But notice from the proof:

$$\frac{1}{u^2} E[R(ux)] = E \left[\frac{R(ux)}{u^2} \right] = E[h(ux)X^2]$$

and for all y , $|h(y)| \leq 1$ as discussed earlier.

Now, since $|h(y)| \leq 1$, we have $E[h(ux)X^2] = \int_{\mathbb{R}} h(ux)X^2 d\mu$

Where the integrand is bounded above by the function x^2 . This is integrable because $\text{Var}(x)$ is finite. This is an assumption of the central limit theorem. If $\text{Var}(x)$ is not finite, the central limit theorem is not always true! The Cauchy distribution is an example of this. Also, since $h(y)$ goes to 0 as y goes to 0, we can do as follows as $u \rightarrow 0$:

$$h(ux) \rightarrow h(0) = 0$$

$$h(uX)X^2 \rightarrow 0$$

So our integral converges pointwise to 0, so all the hypotheses of the dominated convergence theorem are met, so this term that we needed to go to 0 goes to 0, so done.

□

9 Chi squared tests

9.1 Setup and explanation of degrees of freedom

This is where uniqueness of characteristic functions gets dethroned as the hardest thing we prove in this website.

Due to the amount of definitions that will be floating around in this proof, I recommend writing them down to try to follow as you go through, otherwise following this may be difficult.

The problem is defining what a degree of freedom is AND showing that the test statistic $\sum_{r=1}^k \frac{(O_r - E_r)^2}{E_r}$ is well approximated for fairly large n by a distribution that depends ONLY on this number. And in the proof we will better understand what this distribution (That you are taught is a chi squared distribution) really is. We will prove 3 claims that will on their own explain quite well why chi squared tests work and then use them to prove a master theorem which says that the cumulative distribution function converges to this distribution as we add more trials to any fixed model.

NOTE: “Fixed model” means the following:

- Fixed number of cells
- Fixed non-zero probabilities
- Result proven as $n \rightarrow \infty$
- Additional constraints possibly, linear only.

Definition. A random vector is a vector of random variables that are possibly not independent. It may have its own probability density function of multiple variables and it always has its own cumulative distribution function.

Now the setup of a chi squared test is that we have a list of probabilities of several events determined by the null hypothesis. For example, for testing if a dice is fair, this list is $(\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6})$, and in this proof we will call these $(p_1, p_2, p_3, p_4, p_5, p_6)$

Now if we roll the dice 120 times, our probability distribution becomes a lattice in 5D space with the “mean/expected” vector being $(np_1, np_2, \dots, np_k) = (20, 20, 20, 20, 20, 20)$ in this case. In general the lattice is in (k-1)-D space where k is the number of cells in the table, since we have a constraint that the sum of all entries in the table equals a fixed total, which causes the dimension to decrease by 1.

If we have a contingency table then fixing the first r-1 row totals adds r-1 constraints and fixing the last row does nothing as fixing the rest fixes the last one automatically, and similarly we have c-1 constraints for columns and each one decreases the dimension of 1. Since the unconstrained model has rc-1 degrees of freedom as there are rc cells, the constrained model has rc-r-c+1 degrees of freedom which is (r-1)(c-1) if you expand it. We will treat the contingency table case as conditioning on the relevant subspace. We’ll prove it generally for conditioning on linear subspaces subject to some conditions.

Therefore the first step of understanding what a degree of freedom is is done.

9.2 Outline of the proof

From now on I will use x_r to denote the r’th entry of a vector x. Here x will represent proportions.

Now consider our contingency table or list of observed data to be a random vector X and total n and number of cells k. will consider the random vector with general r’th entry $\sqrt{n} \left(\frac{x_r - p_r}{\sqrt{p_r}} \right)$, where we note that the expected value of $\bar{x}_r$ is p_r .

(In the dice example, the number of ones divided by the number of rolls would be $\bar{x}_1$ and the expected value of this is $p_1 = \frac{1}{6}$). This begs the question of why we are specifically considering the vector given by $\sqrt{n} \left(\frac{\bar{x}_r - p_r}{\sqrt{p_r}} \right)$, which is what I call the standardized version of m, or m moved to the standardized world, where m is what I am calling the random vector of the table itself. The answer is because of what happens when you compute the square of its magnitude.

The square of the magnitude is the sum of the squares of its components. This is $\sum_{r=1}^k n \frac{(\bar{x}_r - p_r)^2}{p_r}$. Lets now multiply the numerator and denominator by n to get

$$\sum_{r=1}^k \frac{n^2 (\bar{x}_r - p_r)^2}{np_r} = \sum_{r=1}^k \frac{(n\bar{x}_r - np_r)^2}{np_r} = \sum_{r=1}^k \frac{(O_r - E_r)^2}{E_r}$$

This is exactly the test statistic.

Now a chi squared distribution is defined as follows:

Take a normal distribution in k dimensions where each component of this k-dimensional random vector is $N(0,1)$ and they are all independent, then note that the probability density of this function is rotationally symmetric (This is a property of the normal distribution we discussed earlier and it comes from pythagoras – this property is why it comes up so much). A χ_k^2 distribution is defined as the distribution of the square of the absolute value of this random vector, or the sum of the squares of its components, or the sum of the squares of k $N(0,1)$ variables.

Therefore, we want to investigate $\sqrt{n} \left(\frac{\bar{x}_r - p_r}{\sqrt{p_r}} \right)$ when constrained to lattices in the plane and hope that we can show that it is standard normal so that we get the result we want. To do this, we will show that the probability of all versions of the contingency table approach what a normal distribution would predict, in a precise sense described below. It is not enough to merely show that the total-only constraint version converges to a k-1-dimensional normal and then assert that constraining a k-1 dimensional normal to s-dimensional slices gives an s-dimensional normal, as although this is true, if the actual probability distribution of tables had a small number of bad slices, the whole k-1 dimensional thing could still converge to a standard normal. This is essentially what I did in the old version of the proof, although with some bounding arguments to try to rule out this behavior which may have been enough to derive a contradiction if we had a bad slice for infinitely many n but this approach is cleaner and more robust. Therefore we need to control the probability of each table configuration, as if we do that instead of controlling just the limiting distribution which allows for probabilities of nearby tables to oscillate and average out and generally behave badly, it will be enough. Note that we only consider linear constraints (i.e. constraints to flat regions, not curved regions) in this proof.

Important notation we will use. Take the vector of expected values and the nearest lattice point in the unconstrained version of the distribution (there is one at most 1 count away in each cell) and call this table m^* , and let y be $\left| \sqrt{n} \left(\frac{\bar{x} - p}{\sqrt{p}} \right) \right|$ (the “standardized” vector). Also fix a $0 < \varepsilon < \frac{1}{6}$ (it will be later that we see why the upper bound was $\frac{1}{6}$) and a region R defined by $|x - E| < n^{\frac{1}{2} + \varepsilon}$, and let m be an arbitrary table. Here x is the counts vector.

To clarify m^* , it is an integer table where everything is rounded up or down from the expected counts, m is any arbitrary table. For example, if we roll a dice 100 times the expected counts would be 100/6 in each cell which is not an integer, so m^* could be something like (16, 16, 17, 17, 17, 17) where these are arranged in any order, and how we pick m^* does not really matter but we will have several choices for it in general, but as long as we use an integer that is either the smallest integer greater than or greatest integer less than the expected counts we are fine. m is the table where we record the observed stuff.

Note that this always exists as we can just make a list that is rounded down from the expected and then add 1 to each element and stop when the total is correct.

Then, using this notation, we claim that

1. $p(m \in R) \rightarrow 1$ as $n \rightarrow \infty$
2. $\sup_{m \in R} \left(\left| \frac{p(m)}{p(m^*) e^{-\frac{1}{2}|y|^2}} - 1 \right| \right) \rightarrow 0$ as $n \rightarrow \infty$

These are parts 1 and 2 of the theorem that we will prove to justify chi squared tests.

Note that we are considering the ratio $\frac{p(m)}{p(m^*)e^{-\frac{1}{2}|y|^2}}$ here as the standard normal predicts that the ratio between the probability of an arbitrary table and the probability of the peak table is about $e^{-\frac{1}{2}|y|^2}$

In english, we will show that for large enough n , the probability of any table in a region with probability approaching 1 as n gets large can be made close to the prediction of the standard normal with as narrow of a margin as we want, and also that this is true when conditioning to a lattice.

Note a subtlety here: If we merely showed that $\left| \frac{p(m)}{p(m^*)e^{-\frac{1}{2}|y|^2}} - 1 \right| \rightarrow 0$ for each m this would not be enough. I know this isn't a good example as m depends on n , but the idea is we want uniform convergence instead of pointwise convergence (which we defined and discussed in Level 6.2). Note that we need uniform convergence for this proof because it allows us to pass the limit through the integral at the end, which is why we add in the supremum operator, because we need to show that there is a fixed n where the normal approximation is fairly good everywhere.

At this point, Claims 1 and 2 give us uniform control over the probability of every table in the central region R . Because this control holds table-by-table, we are allowed to impose a linear constraint and reason exactly as in the continuous normal case: inside R , conditioning simply restricts us to a lower-dimensional normal shape (by symmetry, as explained earlier). What is not yet guaranteed is that conditioning cannot concentrate a significant amount of probability into the tails, since Claim 1 only controls the total probability of the tails before conditioning. A third claim is therefore needed to show that the tail bound from Claim 1 still holds after conditioning, so that almost all of the conditional probability remains inside R .

3. Define R_2 by $|x - E| < An^{\frac{1}{2}+\epsilon}$ (where A is a constant that depends on the specific probabilities but not on n which we will explain how to find in the proof of the lemma) which still has probability going to 1 by precisely the same proof as part 1 of the claim. Then the claim is that $R_2 \subseteq R$ for our choice of A and that for any S intersecting R_2 , $p(m \in R \mid m \in S) \rightarrow 1$ as $n \rightarrow \infty$ in the specific case where S is any lattice we constrain to when we are doing a chi squared test: We need this because we only have control inside R by parts 1 and 2 so in principle there could be “bad lattices” where this does not hold without contradicting part 1 provided this happens with probability going to 0 as $n \rightarrow \infty$.

Subtlety: We need it to be the case that the set of possible points in the lattice we constrain to does not have lower dimension than the lattice we are constraining to. What I mean is, if the table was 2 cells that both have 1 and 1, then a constraint like “ $\sqrt{2}\text{cell } 1 + \text{cell } 2 = 1 + \sqrt{2}$ ” seems like we are constraining to 1 dimension, but really the set of tables actually in that constraint is 0 dimensional. Luckily, in all practical cases, it is easy to show that this dimension thing.

Subtlety 2: Note that when constraining we will let n get large but we will only consider n such that it is a multiple of the “least n such that there is a full-dimension set of compatible table” so that the dimensions line up.

If we have an $r * c$ contingency table then we can change it in $(r - 1)(c - 1)$ independent ways and still get a valid contingency table, unless one of the cells is 0, but as we will see later in the proof this is not an issue. We can do this by picking a row number A between 1 and $r-1$ inclusive and a column number b between 1 and $c-1$ inclusive and doing the following:

- Add 1 to the (A,B) and (r,c) cells
- Subtract 1 from the (A,c) and (r,B) cells

This leaves the constraints satisfied and there are $(r-1)(c-1)$ directions we can move in this way, so that is the dimension of the set of possible tables.

Since we control the probability of every single possible table as well as the behavior in every slice in the “almost certain” zone, we can say confidently once we have proven this that chi squared tests work better as n gets large as if n is large enough we will eventually have to go far into the “rejection zone” to run into issues anyways.

Note that in the unconstrained model, $|m^* - E| \leq \sqrt{k}$. This is because there exists a possible table within a distance $\sqrt{k}$ of E and the reason is because we can round everything in E down and then add 1 to each component until the total is correct, and then each component will be within 1 of E and it will be a possible table with distance at most $\sqrt{k}$ from E .

Given this preamble, we aim to prove the following theorem which due to some lore I have nicknamed the Wilson Gerald

Ford theorem.

Theorem. (Wilson Gerald Ford theorem) Let X be a multinomial random vector with n trials and non-zero probabilities and k possible outcomes, expected outcomes O and observed outcomes E . then let $S = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i}$. Suppose now that we consider S conditional on O and E both satisfying a system of no or possibly some linear equations, and that as a result v is the smallest dimensional hyperplane that contains all the possibilities of O under this constraint and $v > 0$. Suppose also that n is appropriate so that v equals $k - 1$ minus the number of independent linear equations we impose O and E both satisfy and that there exists such an n for the given system of equations. Suppose also that the set of expected counts Then as $n \rightarrow \infty$, S converges in distribution to a distribution depending only on v , and not on the probabilities or k .

This theorem implies the normal and contingency table cases. Parameter estimation for binomial case and the “5” rule will be handled separately.

9.3 Stirling's approximation

Lemma 1. (Stirling's approximation) For all positive integers n , $\sqrt{2\pi n} \left(\frac{n}{e}\right)^n < n! \leq \sqrt{2\pi n} \left(\frac{n}{e}\right)^n e^{\frac{1}{12n}}$

Proof. Define $d_n := \log \left(\frac{n! e^n}{n^{n+\frac{1}{2}}} \right) = \log(n!) - \left(n + \frac{1}{2}\right) \log(n) + n$

Then $d_n - d_{n+1} = \log(n!) - \left(n + \frac{1}{2}\right) \log(n) + n - \log((n+1)!) + \left(n + \frac{3}{2}\right) \log(n+1) - n - 1$

$$d_n - d_{n+1} = -\log(n+1) - \left(n + \frac{1}{2}\right) \log(n) + \left(n + \frac{3}{2}\right) \log(n+1) - 1$$

$$d_n - d_{n+1} = -\left(n + \frac{1}{2}\right) \log(n) + \left(n + \frac{1}{2}\right) \log(n+1) - 1 = \left(n + \frac{1}{2}\right) \left(\log\left(\frac{n+1}{n}\right) \right) - 1$$

Now do the substitution $t := \frac{1}{2n+1}$. $n = \frac{1-t}{2t}$ Then $d_n - d_{n+1} = \left(\frac{1}{2} + \frac{1-t}{2t}\right) \log\left(\frac{\frac{1-t}{2t}+1}{\frac{1-t}{2t}}\right) - 1$

$$= \frac{1}{2t} \log\left(\frac{1+t}{1-t}\right) - 1$$

Note that by the log taylor series, $\log(1+t) = t - \frac{t^2}{2} + \frac{t^3}{3} - \frac{t^4}{4} + \dots$ ($|t| < 1$ automatically by how t is defined here so it is valid), and $\log(1-t) = -\left(t + \frac{t^2}{2} + \frac{t^3}{3} + \frac{t^4}{4} + \dots\right)$.

Therefore $\frac{1}{2t} \log\left(\frac{1+t}{1-t}\right) - 1 = \frac{1}{2t} \log(1+t) - \frac{1}{2t} \log(1-t) - 1$

$$\begin{aligned} &= \frac{1}{2t} \left(t - \frac{t^2}{2} + \frac{t^3}{3} + \frac{t^4}{4} + \dots \right) + \frac{1}{2t} \left(t + \frac{t^2}{2} + \frac{t^3}{3} + \frac{t^4}{4} + \dots \right) - 1 \\ &= \frac{1}{2t} \left(2t + \frac{2t^3}{3} + \frac{2t^5}{5} + \frac{2t^7}{7} + \dots \right) - 1 = 1 + \frac{t^2}{3} + \frac{t^4}{5} + \frac{t^6}{7} + \dots - 1 = \frac{t^2}{3} + \frac{t^4}{5} + \frac{t^6}{7} + \dots \end{aligned}$$

t is positive so $d_n - d_{n+1} < \frac{t^2}{3} + \frac{t^4}{5} + \frac{t^6}{7} + \dots = \frac{1}{3} \frac{t^2}{1-t^2}$ (geometric series) $= \frac{1}{3((2n+1)^2-1)}$ by definition of t . By simple algebra this is $\frac{1}{12} \left(\frac{1}{n} - \frac{1}{n+1} \right)$.

Now apply the method of differences/telescoping. $d_1 - d_n = d_1 - d_2 + d_2 - d_3 + \dots + d_{n-1} - d_n$

$$< \frac{1}{12} \left(\frac{1}{1} - \frac{1}{2} \right) + \frac{1}{12} \left(\frac{1}{2} - \frac{1}{3} \right) + \dots + \frac{1}{12} \left(\frac{1}{n-1} - \frac{1}{n} \right) = \frac{1}{12} \left(1 - \frac{1}{n} \right)$$

Note that $d_n - d_{n+1}$ is positive so d is a decreasing sequence, and for all n , d_n is bounded below by $d_1 - \frac{1}{12}$ for all n by the inequality above. A decreasing sequence that is bounded below converges and therefore our sequence d_n converges to a limit which we will call A which is also a strict lower bound for all terms d_n in our decreasing sequence.

Now let $m > n$, then $d_n - d_m < \frac{1}{12} \left(\frac{1}{n} - \frac{1}{n+1} \right) + \frac{1}{12} \left(\frac{1}{n+1} - \frac{1}{n+2} \right) + \dots + \frac{1}{12} \left(\frac{1}{m-1} - \frac{1}{m} \right) = \frac{1}{12} \left(\frac{1}{n} - \frac{1}{m} \right)$. Taking the limit as $m \rightarrow \infty$ we get that $d_n - A \leq \frac{1}{12n}$ (not strict as a sequence always strictly above a value can converge to that value). Therefore $A < d_n \leq A + \frac{1}{12n}$.

Now we will do some magic by working on a seemingly irrelevant integral where A magically pops out.

$$I_n := \int_0^{\frac{\pi}{2}} \sin^n(\theta) d\theta = [-\cos(\theta) \sin^{n-1}(\theta)]_0^{\frac{\pi}{2}} + (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2}(\theta) \cos^2(\theta) d\theta$$

Where we did integration by parts integrating sin and differentiating $\sin^{n-1}(\theta)$ and were careful to not flip any minus signs the wrong way even when we had double negatives to deal with.

$I_n = (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2}(\theta) - \sin^n(\theta) d\theta$ because the bracketed part vanishes to 0 and we have the pythagorean identity $\cos^2(\theta) = 1 - \sin^2(\theta)$. Thus $I_n = (n-1)(I_{n-2} - I_n)$ so $nI_n = (n-1)I_{n-2}$ so we have our recurrence relation $I_n = \frac{n-1}{n}I_{n-2}$ for n at least 2. Since $I_0 = \frac{\pi}{2}$ and $I_1 = 1$ by basic calculus we have the following:

$I_{2n} = \frac{1}{2} \frac{3}{4} \dots \frac{2n-1}{2n} \frac{\pi}{2}$ and $I_{2n+1} = \frac{2}{3} \frac{4}{5} \dots \frac{2n}{2n+1}$ by the recurrence relation. It is also easy to see that I_n is a decrerasing sequence since the functions $\sin^n(\theta)$ are decreasing as n increases as $0 < \sin^n(\theta) < 1$ everywhere in the domain except at $\frac{\pi}{2}$. Therefore $1 \leq \frac{I_{2n}}{I_{2n+1}} \leq \frac{I_{2n-1}}{I_{2n+1}} = 1 + \frac{1}{2n}$ by the recurrence relation and therefore $\frac{I_{2n}}{I_{2n+1}} \rightarrow 1$ and similarly so does $\frac{I_{2n-1}}{I_{2n}}$.

$$I_{2n} = \frac{1}{2} \frac{3}{4} \dots \frac{2n-1}{2n} \frac{\pi}{2} = \frac{1 * 2 * 3 * \dots * 2n}{2 * 2 * 4 * 4 * \dots * 2n * 2n} \frac{\pi}{2} = \frac{(2n)!}{(n!2^n)^2} \frac{\pi}{2}$$

$$I_{2n+1} = \frac{2}{3} \frac{4}{5} \dots \frac{2n}{2n+1} = \frac{2 * 2 * 4 * 4 * \dots * 2n * 2n}{1 * 2 * 3 * \dots * 2n * (2n+1)} = \frac{(n!2^n)^2}{(2n+1)!}$$

Note that when n is large the ratio between $n!$ and $n^{n+\frac{1}{2}}e^{-n+A}$ converges to 1 as the log of the difference converges to 0 by the definition of A and the definition of d_n .

$$\text{Now } \frac{I_{2n}}{I_{2n+1}} = \frac{\frac{(2n)!}{(n!2^n)^2} \frac{\pi}{2}}{\frac{(n!2^n)^2}{(2n+1)!}} = \frac{(2n)!(2n+1)!\pi}{(n!2^n)^4 2} = \pi(2n+1) \left[\frac{((2n)!)^2}{2^{4n+1}(n!)^4} \right]$$

$$\begin{aligned} \frac{((2n)!)^2}{2^{4n+1}(n!)^4} &= \frac{\left(\frac{(2n)!e^{2n}}{(2n)^{2n+\frac{1}{2}}} \right)^2}{2^{4n+1}(n!)^4} \frac{e^{-4n}}{(2n)^{-1-4n}} = \frac{e^{2d_{2n}}}{2^{4n+1} \left(\frac{n!e^n}{n^{n+\frac{1}{2}}} \right)^4} \frac{e^{4n}}{n^{4n+2}} \frac{e^{-4n}}{(2n)^{-1-4n}} = \frac{e^{2d_{2n}}}{2^{4n+1}e^{4d_n}} \frac{1}{n^{4n+2}} \frac{1}{(2n)^{-1-4n}} \\ &= \frac{e^{2d_{2n}}}{2^{4n+1}e^{4d_n}} \frac{1}{n^{4n+2}} \frac{1}{2^{-1-4n}n^{-1-4n}} = \frac{e^{2d_{2n}}}{e^{4d_n}} \frac{1}{n} \end{aligned}$$

As n gets large the ratio between this and $\frac{1}{ne^{2A}} \rightarrow 1$ and thus the limit as n goes to infinity of the ratio $\frac{I_{2n}}{I_{2n+1}} = \pi(2n+1) \left[\frac{((2n)!)^2}{2^{4n+1}(n!)^4} \right]$ (which we know is 1) is exactly the limit of $\pi(2n+1) \frac{1}{ne^{2A}} = \frac{2\pi}{e^{2A}}$, and therefore A must be $\log(\sqrt{2\pi})$. I told you this would me magic!

Thus, we are about to get the lemma.

$$\begin{aligned} A &< d_n \leq A + \frac{1}{12n} \\ \log(\sqrt{2\pi}) &< \log\left(\frac{n!e^n}{n^{n+\frac{1}{2}}}\right) \leq \log(\sqrt{2\pi}) + \frac{1}{12n} \\ \sqrt{2\pi} &< \frac{n!e^n}{n^{n+\frac{1}{2}}} \leq \sqrt{2\pi}e^{\frac{1}{12n}} \\ \sqrt{2\pi n} &< \frac{n!e^n}{n^n} \leq \sqrt{2\pi n}e^{\frac{1}{12n}} \\ \sqrt{2\pi n}\left(\frac{n}{e}\right)^n &< n! \leq \sqrt{2\pi n}\left(\frac{n}{e}\right)^n e^{\frac{1}{12n}} \end{aligned}$$

□

Remark. $\sqrt{2\pi n} \left(\frac{n}{e}\right)^n e^{\frac{1}{12n}}$ is far closer to $n!$ in practice than the lower bound, in fact it is possible to show that $\sqrt{2\pi n} \left(\frac{n}{e}\right)^n e^{\frac{1}{12n+1}}$ is a lower bound and strict inequalities are true for both bounds but we will not prove this as although it is not too difficult it is quite long and technical and the bound given above is sufficient for the proof we will do.

9.4 Proof in “k-1 df” case

Proof of part 1 of the claim:

We will work on the unconstrained model for now since the bounds we will get on the probabilities will allow us to condition onto sub-lattices later.

Suppose $|x - E| \geq n^{\frac{1}{2}+\varepsilon}$ but for all r from 0 to k , $|O_r - E_r| < \frac{n^{\frac{1}{2}+\varepsilon}}{\sqrt{k}}$. Then we would have $|x - E| < n^{\frac{1}{2}+\varepsilon}$ by pythagoras ($|x - E| = \sqrt{\sum_{r=1}^k (O_r - E_r)^2} < \sqrt{\sum_{r=1}^k \frac{n^{1+2\varepsilon}}{k}} = \sqrt{n^{1+2\varepsilon}} = n^{\frac{1}{2}+\varepsilon}$). Hence $|x - E| \geq n^{\frac{1}{2}+\varepsilon}$ implies that there is at least one r with $|O_r - E_r| \geq \frac{n^{\frac{1}{2}+\varepsilon}}{\sqrt{k}}$. But the r 'th entry of the vector is on its own is distributed like a binomial distribution (minus the mean) so its standard deviation is proportional to $\sqrt{n}$, so the number of standard deviations away the r 'th entry of the vector would have to be from its value is proportional to n^ε which goes to infinity (slowly) as n goes to infinity.

We want to show that the probability of the r 'th component being out by that much goes to 0, as it will then follow that the probability of any of the k components being out by that much is 0 since it is a fixed number (k) of separate events each with probability going to 0. Well, it *seems* obvious that being out by an unbounded number of standard deviations has probability approaching 0 but we want to show this rigorously.

Therefore we will prove an easy inequality (due to markov) that says that $P(y \geq a) \leq \frac{E(y)}{a}$ where y is any non-negative distribution and $a > 0$.

The proof is that $E(y) \geq E\left(\begin{matrix} y & y \geq a \\ 0 & y < a \end{matrix}\right) \geq E\left(\begin{matrix} a & y \geq a \\ 0 & y < a \end{matrix}\right) = aP(y \geq a)$ so rearranging gives the desired result.

We prove another easy inequality (due to chebyshev) that says that for any distribution x we have that

$$P(|x - E(x)| \geq t) \leq \frac{Var(x)}{t^2}$$

The proof is simple: Just note $P(|x - E(x)|^2 \geq t^2) \leq \frac{Var(x)}{t^2}$ is equivalent to what we are claiming but it is just the above inequality applied to $|x - E(x)|^2$.

Therefore, since $Var(O_r) = np_r(1 - p_r) < n$, taking $t = \frac{n^{\frac{1}{2}+\varepsilon}}{\sqrt{k}}$. Then the inequality above gives that

$$P(|O_r - E_r| \geq t) \leq \frac{Var(x)}{t^2} < \frac{kn}{n^{1+2\varepsilon}} = kn^{-2\varepsilon} \rightarrow 0$$

Note that from now on I use m_r, O_r interchangeably to mean the same thing.

Proof of part 2 of the claim:

We need to better understand the probability of a particular table in the unconstrained model. The idea is we can get it in terms of factorials then apply lemma 1.

Suppose the total is n and the counts are $m_1, m_2, \dots, m_k$. Then we will argue as follows:

We do n trials and the number of ways to interchange the order of these n trials is $n!$, however for each possible outcome, the order of those is irrelevant (For example, if you roll 120 dice, then we can change the order that we got the 1's in 20! Different ways without changing the overall order). Therefore the number of possible ways to order the trials without distinguishing the same outcome is exactly $\frac{n!}{m_1!m_2!\dots m_k!}$, and the probability of one particular ordering is $p_1^{m_1}p_2^{m_2}\dots p_k^{m_k}$. Therefore the exact probability of a particular table is $\frac{n!}{m_1!m_2!\dots m_k!}p_1^{m_1}p_2^{m_2}\dots p_k^{m_k}$.

Taking logs of lemma 1 gives $\log(m!) = (m + \frac{1}{2}) \log(m) - m + \frac{1}{2} \log(2\pi) + r_m$ with $0 < r_m < \frac{1}{12m}$.

Now set $h_i := m_i - m_i^* = O_i - E_i + O(1)$. Therefore in the standardized world, $y_i = \frac{h_i}{\sqrt{np_i}} + O(n^{-\frac{1}{2}})$.

Here when I use say $O(f(x))$ I mean there exists a constant C not depending on n or m such that for all m in R the thing we are saying is $O(f(x))$ is less than $Cf(x)$ in absolute value.

$$\log(P(m)) = \log(n!) - \sum_{i=1}^k \log(m_i!) + \sum_{i=1}^k m_i \log(p_i)$$

By lemma 1,

$$\begin{aligned} \log(P(m)) &= \left(n + \frac{1}{2}\right) \log(n) - n + \frac{1}{2} \log(2\pi) + r_n - \sum_{i=1}^k \left(m_i + \frac{1}{2}\right) \log(m_i) - m_i + \frac{1}{2} \log(2\pi) + r_{m_i} + \sum_{i=1}^k m_i \log(p_i) \\ \log(P(m^*)) &= \left(n + \frac{1}{2}\right) \log(n) - n + \frac{1}{2} \log(2\pi) + r_n - \sum_{i=1}^k \left(m_i^* + \frac{1}{2}\right) \log(m_i^*) - m_i^* + \frac{1}{2} \log(2\pi) + r_{m_i^*} + \sum_{i=1}^k m_i^* \log(p_i) \end{aligned}$$

Then

$$\log(P(m)) - \log(P(m^*)) = - \sum_{i=1}^k \left(m_i + \frac{1}{2}\right) \log(m_i) - h_i + r_{m_i} + \sum_{i=1}^k h_i \log(p_i) + \sum_{i=1}^k \left(m_i^* + \frac{1}{2}\right) \log(m_i^*) + r_{m_i^*}$$

Then the remainder terms contribute $\left| \sum_{i=1}^k r_{m_i} - r_{m_i^*} \right| \leq \sum_{i=1}^k \frac{1}{12} \left| \frac{1}{m_i} - \frac{1}{m_i^*} \right|$, the inequality follows from the same method of differences argument as in lemma 1. Note that we are proving a claim about the region R and therefore m_i is not 0 and we can divide by it.

If $n > \frac{8}{p_i^3}$ for all i and we are in R, $m_i > m_i^* - n^{\frac{2}{3}} > \frac{1}{2} np_i$ which is a constant times n. Therefore, if n is large enough, $\left| \sum_{i=1}^k r_{m_i} - r_{m_i^*} \right| \leq \sum_{i=1}^k \frac{1}{12} \left| \frac{h_i}{m_i m_i^*} \right| \leq C \sum_{i=1}^k \left| \frac{h_i}{n^{\frac{1}{2}}} \right|$ for some fixed C. This is therefore $O(n^{\varepsilon - \frac{3}{2}})$.

$$m_i = m_i^* (1 + u_i) \text{ with } u_i = \frac{h_i}{m_i^*} = \frac{O(n^{\frac{1}{2} + \varepsilon})}{np_i + O(1)} = O(n^{\varepsilon - \frac{1}{2}})$$

Write

$$\Delta(m) := \log(P(m)) - \log(P(m^*)) = \sum_{i=1}^k - \left(m_i + \frac{1}{2}\right) \log(m_i) + h_i + h_i \log(p_i) + \left(m_i^* + \frac{1}{2}\right) \log(m_i^*) + O(n^{\varepsilon - \frac{3}{2}})$$

Write $m_i = m_i^* (1 + u_i)$ so $u_i := \frac{h_i}{m_i^*} < \frac{n^{\frac{1}{2} + \varepsilon}}{np_i + O(1)} = O(n^{\varepsilon - \frac{1}{2}})$.

We note that $\sum h_i = 0$ so

$$\Delta(m) = \sum_{i=1}^k - \left(m_i + \frac{1}{2}\right) \log(m_i) + h_i \log(p_i) + \left(m_i^* + \frac{1}{2}\right) \log(m_i^*) + O(n^{\varepsilon - \frac{3}{2}})$$

Now $(m_i + \frac{1}{2}) \log(m_i) = (m_i^* + h_i + \frac{1}{2}) (\log(m_i^*) + \log(1 + u_i))$

Therefore

$$\Delta(m) = \sum_{i=1}^k - \left(m_i^* + h_i + \frac{1}{2}\right) (\log(m_i^*) + \log(1 + u_i)) + h_i \log(p_i) + \left(m_i^* + \frac{1}{2}\right) \log(m_i^*) + O(n^{\varepsilon - \frac{3}{2}})$$

$$\Delta(m) = \sum_{i=1}^k -m_i^* \log(m_i^*) - h_i \log(m_i^*) - \frac{1}{2} \log(m_i^*) - m_i^* \log(1 + u_i) - h_i \log(1 + u_i) - \frac{1}{2} \log(1 + u_i) + h_i \log(p_i) + \left(m_i^* + \frac{1}{2}\right) \log(m_i^*) + O(n^{\varepsilon - \frac{3}{2}})$$

$$\Delta(m) = \sum_{i=1}^k -h_i \log(m_i^*) - m_i^* \log(1+u_i) - h_i \log(1+u_i) - \frac{1}{2} \log(1+u_i) + h_i \log(p_i) + O(n^{\varepsilon - \frac{3}{2}})$$

$$\Delta(m) = \sum_{i=1}^k -h_i \log(m_i^*) - m_i \log(1+u_i) - \frac{1}{2} \log(1+u_i) + h_i \log(p_i) + O(n^{\varepsilon - \frac{3}{2}})$$

$$h_i (\log(m_i^*) - \log(p_i)) = h_i \left(\log\left(\frac{m_i^*}{p_i}\right) \right) = h_i (\log(n + O(1))) = h_i (\log(n) + O\left(\frac{1}{n}\right))$$

$$\sum_{i=1}^k h_i (\log(m_i^*) - \log(p_i)) = \sum_{i=1}^k h_i \left(O\left(\frac{1}{n}\right) \right) = O(n^{\varepsilon - \frac{1}{2}})$$

This is because $\sum h_i = 0$

Then

$$\Delta(m) = \sum_{i=1}^k -m_i^* \log(1+u_i) - h_i \log(1+u_i) - \frac{1}{2} \log(1+u_i) + O(n^{\varepsilon - \frac{1}{2}})$$

$$\Delta(m) = \sum_{i=1}^k -m_i \log(1+u_i) - \frac{1}{2} \log(1+u_i) + O(n^{\varepsilon - \frac{1}{2}})$$

$$R_3(x) := \log(1+x) - x + \frac{x^2}{2}$$

$$\Delta(m) = \sum_{i=1}^k -m_i \left(u_i - \frac{1}{2} u_i^2 + R_3(u_i) \right) - \frac{1}{2} u_i + \frac{1}{4} u_i^2 - \frac{1}{2} R_3(u_i) + O(n^{\varepsilon - \frac{1}{2}})$$

$$\Delta(m) = \sum_{i=1}^k -m_i \left(u_i - \frac{1}{2} u_i^2 + R_3(u_i) \right) + \sum_{i=1}^k -\frac{1}{2} u_i + \frac{1}{4} u_i^2 - \frac{1}{2} R_3(u_i) + O(n^{\varepsilon - \frac{1}{2}})$$

Note that by definition $h_i = m_i^* u_i$ and $m_i = m_i^* (1+u_i)$ so $m_i u_i = m_i^* (1+u_i) u_i = h_i + m_i^* u_i^2$

Therefore since $\sum h_i = 0$ we get that

$$\Delta(m) = \sum_{i=1}^k -m_i^* u_i^2 + \frac{1}{2} m_i u_i^2 - m_i R_3(u_i) + \sum_{i=1}^k -\frac{1}{2} u_i + \frac{1}{4} u_i^2 - \frac{1}{2} R_3(u_i) + O(n^{\varepsilon - \frac{1}{2}})$$

$$\Delta(m) = \sum_{i=1}^k -\frac{1}{2} m_i^* u_i^2 + \frac{1}{2} h_i u_i^2 - m_i R_3(u_i) + \sum_{i=1}^k -\frac{1}{2} u_i + \frac{1}{4} u_i^2 - \frac{1}{2} R_3(u_i) + O(n^{\varepsilon - \frac{1}{2}})$$

Now use the fact that $u_i = \frac{h_i}{m_i^*}$ to get

$$\Delta(m) = \sum_{i=1}^k -\frac{1}{2} \frac{h_i^2}{m_i^*} + \frac{1}{2} \frac{h_i^3}{(m_i^*)^2} - m_i R_3(u_i) + \sum_{i=1}^k -\frac{1}{2} u_i + \frac{1}{4} u_i^2 - \frac{1}{2} R_3(u_i) + O(n^{\varepsilon - \frac{1}{2}})$$

$$\Delta(m) = \sum_{i=1}^k -\frac{1}{2} \frac{h_i^2}{m_i^*} + \frac{1}{2} \frac{h_i^3}{(m_i^*)^2} - m_i R_3(u_i) - \sum_{i=1}^k \frac{1}{2} R_3(u_i) + O(n^{\varepsilon - \frac{1}{2}})$$

Because $\sum u_i = \sum O(n^{\varepsilon - \frac{1}{2}}) = O(n^{\varepsilon - \frac{1}{2}})$ since the sum is finite and $\sum u_i^2$ gets even smaller, in fact it is $O(n^{2\varepsilon - 1})$ as n gets large.

Now $\frac{h_i^3}{(m_i^*)^2} = \frac{O(n^{\frac{3}{2} + 3\varepsilon})}{O(n^2)}$ where crucially for large enough n $(m_i^*)^2 \geq Cn^2$ for some positive C which allows us to say that its reciprocal is $O(n^{-2})$. It is here we use the fact that $\varepsilon < \frac{1}{6}$, as we now have

$$\Delta(m) = \sum_{i=1}^k -\frac{1}{2} \frac{h_i^2}{m_i^*} - m_i R_3(u_i) - \sum_{i=1}^k \frac{1}{2} R_3(u_i) + O(n^{3\varepsilon - \frac{1}{2}})$$

Now by Taylor's theorem, $R_3(u_i)$ from how it was defined is $O(|u_i|^3)$ where the constant in the big O notation is fixed for large n as the u_i 's tend to 0.

Recall that $u_i = O(n^{\varepsilon - \frac{1}{2}})$ and $m_i = O(n)$ so $R_3(u_i) = O(n^{3\varepsilon - \frac{3}{2}})$ and thus so is $\sum_{i=1}^k \frac{1}{2} R_3(u_i)$, and also $m_i R_3(u_i) = O(n^{3\varepsilon - \frac{1}{2}})$ so at last we have

$$\Delta(m) = \sum_{i=1}^k -\frac{1}{2} \frac{h_i^2}{m_i^*} + O(n^{3\varepsilon - \frac{1}{2}})$$

The last step is

$$\begin{aligned} \Delta(m) &= \sum_{i=1}^k -\frac{1}{2} \frac{h_i^2}{E_i + O(1)} + O(n^{3\varepsilon - \frac{1}{2}}) \\ \Delta(m) &= \sum_{i=1}^k -\frac{1}{2} \frac{h_i^2}{E_i} \left(1 + O\left(\frac{1}{n}\right)\right) + O(n^{3\varepsilon - \frac{1}{2}}) \\ \Delta(m) &= \sum_{i=1}^k -\frac{1}{2} \frac{h_i^2}{E_i} + \sum_{i=1}^k -\frac{1}{2} \frac{h_i^2}{E_i} O\left(\frac{1}{n}\right) + O(n^{3\varepsilon - \frac{1}{2}}) \\ \Delta(m) &= \sum_{i=1}^k -\frac{1}{2} \frac{h_i^2}{E_i} + O(n^{3\varepsilon - \frac{1}{2}}) \end{aligned}$$

Since that term was $O(n^{2\varepsilon - 1})$.

Now we are basically there because:

$$\Delta(m) = \log(P(m)) - \log(P(m^*))$$

Note that h_i is $m_i - m_i^*$ and y_i was defined using $m_i - E_i$, however when we identify $|y|^2$ and $\frac{h_i^2}{E_i}$ we note that the difference in question is $O\left(\frac{1}{\sqrt{n}}\right)$ by how we did the standardization. Therefore it is valid to say that

$$\log(P(m)) - \log(P(m^*)) = -\frac{1}{2} |y|^2 + O(n^{3\varepsilon - \frac{1}{2}})$$

Undoing logs gives

$$\frac{P(m)}{P(m^*)} = e^{-\frac{1}{2} |y|^2} T$$

Where T goes to 1 as n goes to infinity for each fixed set of p values and regardless of m provided m is in R , so done with part 2 of the claim.

9.5 Extension to contingency table case

Lemma 2. There exists constants $c, C > 0$ such that $\sup_{m \notin R} p(m) \leq Cp(m^*) e^{-cn^{2\varepsilon}}$ for sufficiently large n . Assume the probabilities are fixed and non-zero and the number of cells k is fixed.

Proof. We first show the simple inequality that says $(1+t) \log(1+t) \geq t + \frac{t^2}{3}$ for all $t \in [-\frac{1}{2}, \frac{1}{2}]$.

Consider $(1+t) \log(1+t) - t - \frac{t^2}{3}$. The derivative of this is $\log(1+t) - \frac{2t}{3}$ and the second derivative is $\frac{1}{1+t} - \frac{2}{3}$ which in the range we care about is always non-negative. Notice that the derivative at 0 is 0 so we have a graph with positive curvature that flattens at $t=0$ where the value is 0, and thus the graph is non-negative in the range we care about so we finished the inequality.

Define a probability vector $q := (q_1, q_2, \dots, q_k)$, $\sum q_r = 1$. Define $t_r := \frac{q_r}{p_r} - 1$.

Define $D(q|p) := \sum_{r=1}^k q_r \log\left(\frac{q_r}{p_r}\right) = \sum_{r=1}^k p_r (1+t_r) \log(1+t_r)$. Using the inequality above we have the bound $D(q|p) \geq \sum_{r=1}^k p_r \left(t_r + \frac{t_r^2}{3}\right)$ whenever all t_r 's are between $-\frac{1}{2}$ and $+\frac{1}{2}$ - we will handle the case where this is not true separately.

$$D(q|p) \geq \sum_{r=1}^k p_r t_r + \frac{1}{3} \sum_{r=1}^k p_r t_r^2 = \sum_{r=1}^k q_r - p_r + \frac{1}{3} \sum_{r=1}^k p_r t_r^2 = \frac{1}{3} \sum_{r=1}^k p_r t_r^2 = \frac{1}{3} \sum_{r=1}^k \frac{(q_r - p_r)^2}{p_r}$$

Given a table m set $q_r = \frac{m_r}{n}$, i.e. the observed probability, this is the q we will use for the remainder of the proof. The standardized vector in components is given by

$$y_r = \sqrt{n} \frac{q_r - p_r}{\sqrt{p_r}} = \frac{O_r - E_r}{\sqrt{np_r}}$$

Recall that from this we derived

$$|y|^2 = \sum_{r=1}^k \frac{(O_r - E_r)^2}{E_r}$$

and also it is the case that

$$|m - E|^2 = \sum_{r=1}^k (O_r - E_r)^2$$

Therefore

$$|y|^2 = \sum_{r=1}^k \frac{(O_r - E_r)^2}{E_r} \geq \frac{1}{\max_r E_r} \sum_{r=1}^k (O_r - E_r)^2 = \frac{|m - E|^2}{\max_r E_r}$$

By definition of R , if $m \notin R$, $|y|^2 \geq \frac{n^{1+2\varepsilon}}{\max_r E_r} = \frac{n^{2\varepsilon}}{p_{max}}$. Here p_{max} is the largest probability.

In the case where all t_r 's are between $-\frac{1}{2}$ and $+\frac{1}{2}$ we had $D(q \mid p) \geq \frac{1}{3} \sum_{r=1}^k \frac{(q_r - p_r)^2}{p_r} = \frac{1}{3n} \sum_{r=1}^k \frac{(O_r - E_r)^2}{E_r} = \frac{|y|^2}{3n}$ so $nD(q \mid p) \geq \frac{|y|^2}{3}$ so by the previous deduction $nD(q \mid p) \geq \frac{n^{2\varepsilon}}{3p_{max}}$.

If for some r , $|t_r| > \frac{1}{2}$ then the set of possible q vectors (including cases where some are 0) satisfying $|t_r| \geq \frac{1}{2}$ is bounded and since it contains all its boundary points it is closed, and trivially it does not contain p . The function $D(q \mid p)$ for fixed p is continuous on K (we say $0 \log(0) = 0$ for the case when one of the q 's is 0 as that is the limit of $x \log(x)$ as x goes to 0), it must be the case (by the technical results document, extreme value theorem) that $D(q \mid p)$ attains a minimum value on K d_0 somewhere in K . Note that $d_0 > 0$: The proof of this is as follows.

For all positive u , $\log(u) \leq u - 1$ because those curves are tangent at $u=1$ and $\log$ has negative curvature everywhere (differentiate it twice). Therefore $\log\left(\frac{p_r}{q_r}\right) \leq \frac{p_r}{q_r} - 1$ so $q_r \log\left(\frac{p_r}{q_r}\right) \leq p_r - q_r$, therefore $q_r \log\left(\frac{q_r}{p_r}\right) \geq q_r - p_r$, so $\sum_{r=1}^k q_r \log\left(\frac{q_r}{p_r}\right) \geq \sum_{r=1}^k q_r - p_r = 0$.

To make the inequality strict note that $\log(u) = u - 1$ only happens when $u = 1$ so if p and q differ for any r then that term in the sum becomes strictly positive and the rest are non-negative so the inequality becomes strict, hence the inequality is strict whenever q and p are not the same, so certainly everywhere in K .

Therefore everywhere in K , $nD(q \mid p) \geq nd_0$.

Note that as explained earlier the number of ways to get a certain table by doing a bunch of trials is exactly $\frac{n!}{\prod m_r!} \prod p_r^{m_r}$, and $\prod p_r^{m_r} = \exp(n \sum q_r \log(p_r))$ where $\exp(x)$ means e^x . Now define another function $H(q) := -\sum q_r \log(q_r)$ with $0 \log(0)$ defined as 0.

Now $\frac{n!}{\prod m_r!}$ is the number of ways to rearrange the order of trials to get the same table, and similarly the number of orderings when considering the $\prod_{r=1}^k m_r^{m_r}$ coefficient in the expansion of $n^n = (m_1 + m_2 + \dots + m_k)^n$. Therefore that coefficient is exactly $\frac{n!}{\prod m_r!}$ so we have the inequality $\frac{n!}{\prod m_r!} \leq \frac{n^n}{\prod_{r=1}^k m_r^{m_r}}$ so

$$p(m) = \frac{n!}{\prod m_r!} \prod p_r^{m_r} \leq \prod p_r^{m_r} \frac{n^n}{\prod_{r=1}^k m_r^{m_r}}$$

Taking logs gives

$$\begin{aligned} \log(p(m)) &\leq n \log(n) - \sum m_r \log(m_r) + \sum m_r \log(p_r) \\ \log(p(m)) &\leq n \log(n) - \sum n q_r \log(q_r) - \sum n q_r \log(n) + \sum m_r \log(p_r) \\ \log(p(m)) &\leq -\sum n q_r \log(q_r) + \sum n q_r \log(p_r) \end{aligned}$$

$$\log(p(m)) \leq n \sum (q_r) \log\left(\frac{p_r}{q_r}\right) = -nD(q||p)$$

Thus $p(m) \leq e^{-nD(q||p)} \leq e^{-nd_0}$

Now we apply lemma 1 and take logs of both sides of the formula for probabilities of particular unconstrained tables.

$$\log(p(m)) = \log(n!) - \sum_{r=1}^k \log(m_r!) + \sum_{r=1}^k m_r \log(p_r)$$

$$\log(p(m)) = n \log(n) - n + \frac{1}{2} \log(2n\pi) + R_n - \sum_{r=1}^k \left[m_r \log(m_r) - m_r + \frac{1}{2} \log(2m_r\pi) + R_{m_r} \right] + \sum_{r=1}^k m_r \log(p_r)$$

Where $0 < R_n < \frac{1}{12n}$ and $0 < R_{m_r} < \frac{1}{12m_r} \leq \frac{1}{12np_{min}}$, therefore

$$\log(p(m)) = n \log(n) - n + \frac{1}{2} \log(2n\pi) - \sum_{r=1}^k \left[m_r \log(m_r) - m_r + \frac{1}{2} \log(2m_r\pi) \right] + \sum_{r=1}^k m_r \log(p_r) + R$$

Where R goes to 0 as n goes to infinity.

Simplifying,

$$\log(p(m)) = n \log(n) + \frac{1}{2} \log(2n\pi) - \sum_{r=1}^k \left[m_r \log(m_r) + \frac{1}{2} \log(2m_r\pi) \right] + \sum_{r=1}^k m_r \log(p_r) + R$$

Because n is the sum of the m_r 's

$$\log(p(m)) = n \log(n) + \frac{1}{2} \log(2n\pi) - \sum_{r=1}^k [m_r \log(m_r)] - \frac{1}{2} \sum_{r=1}^k [\log(2m_r\pi)] + \sum_{r=1}^k m_r \log(p_r) + R$$

$$\log(p(m)) = \frac{1}{2} \log(2n\pi) - \sum_{r=1}^k [m_r [\log(q_r)]] - \frac{1}{2} \sum_{r=1}^k [\log(2m_r\pi)] + \sum_{r=1}^k m_r \log(p_r) + R$$

$$\log(p(m)) = \frac{1}{2} \log(2n\pi) - \sum_{r=1}^k \left[m_r \left[\log\left(\frac{q_r}{p_r}\right) \right] \right] - \frac{1}{2} \sum_{r=1}^k [\log(2m_r\pi)] + R$$

$$\log(p(m)) = \frac{1}{2} \log(2n\pi) - n \sum_{r=1}^k \left[q_r \left[\log\left(\frac{q_r}{p_r}\right) \right] \right] - \frac{1}{2} \sum_{r=1}^k [\log(2m_r\pi)] + R$$

$$\log(p(m)) = \frac{1}{2} \log(2n\pi) - nD(q||p) - \frac{1}{2} \sum_{r=1}^k [\log(2m_r\pi)] + R$$

$$\log(p(m)) = \frac{1-k}{2} \log(n) - nD(q||p) - \frac{1}{2} \sum_{r=1}^k [\log(q_r)] - \frac{k-1}{2} \log(2\pi) + R$$

Note that $\frac{k-1}{2} \log(2\pi)$ is a fixed constant here.

Note that $|m^* - E| \leq \sqrt{k}$ from earlier, therefore $|q^* - p| \leq \frac{\sqrt{k}}{n}$.

By the technical results document (Taylor's theorem),

$$\log\left(\frac{q_r}{p_r}\right) = \log\left(1 + \frac{q_r - p_r}{p_r}\right) = \frac{q_r - p_r}{p_r} - \frac{1}{2} \left(\frac{q_r - p_r}{p_r}\right)^2 + O\left(\left(\frac{q_r - p_r}{p_r}\right)^3\right)$$

But then

$$q_r \log\left(\frac{q_r}{p_r}\right) = (q_r) \left[\frac{q_r - p_r}{p_r} - \frac{1}{2} \left(\frac{q_r - p_r}{p_r}\right)^2 + O\left(\left(\frac{q_r - p_r}{p_r}\right)^3\right) \right]$$

Now set $\delta_r := q_r^* - p_r$, and we know from what we did a few lines ago that $\delta_r = O\left(\frac{1}{n}\right)$.

$$\begin{aligned} q_r^* \log\left(\frac{q_r^*}{p_r}\right) &= (p_r + \delta_r) \left[\frac{\delta_r}{p_r} - \frac{1}{2} \left(\frac{\delta_r}{p_r}\right)^2 + O\left(\left(\frac{\delta_r}{p_r}\right)^3\right) \right] \\ &= \delta_r - \frac{1}{2} \frac{\delta_r^2}{p_r} + \frac{\delta_r^2}{p_r} + O(\delta_r^3) = \delta_r + \frac{1}{2} \frac{\delta_r^2}{p_r} + O(\delta_r^3) \end{aligned}$$

The p 's don't affect the error term growth rate because the p 's are fixed and non-zero. Note however that when we sum this over r to get $D(q^*||p)$, the δ_r terms cancel since those add to 0 as q^* and p add up to 1, and since $\delta_r = O\left(\frac{1}{n}\right)$, $D(q^*||p) = O\left(\frac{1}{n^2}\right)$ and therefore $nD(q^*||p) = O\left(\frac{1}{n}\right)$.

Therefore we have

$$\log(p(m)) = \frac{1-k}{2} \log(n) - nD(q||p) - \frac{1}{2} \sum_{r=1}^k [\log(q_r)] + O(1)$$

From earlier, and

$$\log(p(m^*)) = \frac{1-k}{2} \log(n) - \frac{1}{2} \sum_{r=1}^k [\log(q_r^*)] + O(1)$$

Because the $nD(q||p)$ term is negligible in the m^* case.

Since the terms $\frac{1}{2} \sum_{r=1}^k [\log(q_r^*)] + O(1)$ stay bounded, we reduce to

$$\log(p(m^*)) = \frac{1-k}{2} \log(n) + O(1)$$

Therefore

$$\log\left(\frac{p(m)}{p(m^*)}\right) = -nD(q||p) - \frac{1}{2} \sum_{r=1}^k [\log(q_r)] + O(1)$$

Therefore we just need to bound $\sum_{r=1}^k [\log(q_r)]$. Outside of K , this is fine, as each q is bounded between $\frac{1}{2}p_r$ and $\frac{3}{2}p_r$ which is away from 0. Therefore we have outside of K that

$$\log\left(\frac{p(m)}{p(m^*)}\right) = -nD(q||p) + O(1)$$

Therefore, if n is large enough, $\frac{p(m)}{p(m^*)} \leq Ce^{-nD(q||p)}$ outside K for some C provided n is large enough.

Therefore, because outside K we established earlier that $nD(q||p) \geq \frac{n^{2\varepsilon}}{3p_{max}}$, it follows that $\frac{p(m)}{p(m^*)} \leq Ce^{-cn^{2\varepsilon}}$ for some C, c independent of n and m when n is sufficiently large so the lemma is proved outside K .

Since $\log(p(m^*)) = \frac{1-k}{2} \log(n) + O(1)$, it means that for n large enough, there exists positive constants A and B such that

$$An^{\frac{1-k}{2}} < p(m^*) < Bn^{\frac{1-k}{2}}$$

Inside K , we established that $p(m) \leq e^{-d_0 n}$ so $\frac{p(m)}{p(m^*)} \leq C_2 n^{\frac{k-1}{2}} e^{-d_0 n} \leq Ce^{-\min(\frac{d_0}{2}, c)n} \leq Ce^{-\min(\frac{d_0}{2}, c)n^{2\varepsilon}}$ for some C_2 (because $\varepsilon < \frac{1}{2}$) and any large enough n and c as earlier. The second inequality is because $\frac{C_2}{C} n^{\frac{k-1}{2}}$ eventually is beaten by $e^{\frac{d_0}{2}n}$ and therefore certainly beaten by $\max\left(e^{\frac{d_0}{2}n}, e^{(d_0-c)n}\right)$ (By the ‘‘Exponentials beat polynomials’’ principle). This proves the lemma in all cases.

□

Proof of part 3 of the claim:

Proof. Pick A from the statement of the claim to be $\sqrt{cp_{\min}}$ where c is as in the lemma 2 statement and $p_{\min}$ is the smallest probability in the table. Note that this is < 1 because c is actually just $\leq \frac{1}{3p_{\max}}$ by our proof of lemma 2 so we have $\sqrt{\frac{p_{\min}}{3p_{\max}}}$ which is clearly less than 1. Thus, $R_2 \subseteq R$.

Define R^C as the outside of R. Pick $m_0 \in S \cap R_2$, then $P(m \notin R | m \in S) = \frac{\sum_{m \in S \cap R^C} p(m)}{\sum_{m \in S} p(m)} \leq \frac{\sum_{m \in S \cap R^C} p(m)}{p(m_0)}$. Let $|S|$ be the number of lattice points in S, dependent on n. Then $\sum_{m \in S \cap R^C} p(m) \leq |S| \sup_{m \notin R} p(m)$. By lemma 2, if n is large enough, $\sup_{m \notin R} p(m) \leq Cp(m^*)e^{-cn^{2\varepsilon}}$. By part 2, if n is large enough, since $m_0 \in R$ we can force $p(m_0) \geq \frac{1}{2}p(m^*)e^{-\frac{1}{2}|y_0|^2}$ where y_0 is the vector m_0 moved to the standardized world.

Note that normally in mathematics, saying A|B requires B to have non-zero probability - however in this case we are fine because we are conditioning on something with probability that approaches 0 but is never 0.

Now since the standardization is $y_i = \frac{m_i - E_i}{\sqrt{np_i}}$ we have

$$|y_0|^2 = \sum_{i=1}^k \frac{((m_0)_i - E_i)^2}{E_i} \leq \frac{1}{np_{\min}} \sum_{i=1}^k ((m_0)_i - E_i)^2 = \frac{|m_0 - E|^2}{np_{\min}}$$

Since $m_0 \in R_2$, $|y_0|^2 \leq \frac{1}{np_{\min}} cp_{\min} n^{1+2\varepsilon}$ by how R_2 is defined, and this is just $|y_0|^2 \leq cp_{\min} \frac{n^{2\varepsilon}}{p_{\min}}$. Therefore we have that $p(m_0) \geq \frac{1}{2}p(m^*)e^{-\frac{1}{2}cn^{2\varepsilon}}$. Going back to the original inequality and putting in all our results, we get the following:

$$P(m \notin R | m \in S) \leq \frac{\sum_{m \in S \cap R^C} p(m)}{p(m_0)} \leq |S| \frac{Cp(m^*)e^{-cn^{2\varepsilon}}}{\frac{1}{2}p(m^*)e^{-\frac{1}{2}cn^{2\varepsilon}}} = 2C|S|e^{-\frac{1}{2}cn^{2\varepsilon}}$$

Now there are n+1 possibilities for each cell and k cells so we get the trivial upper bound which says $|S| \leq (n+1)^k$. Therefore,

$$P(m \notin R | m \in S) \leq 2C(n+1)^k e^{-\frac{1}{2}cn^{2\varepsilon}}$$

Again exponentials beat polynomials so the above goes to 0, completing the proof of the proposition. So done. □

Proof that this implies chi squared tests work:

Sketch of proof. The idea is that the bounds above let you do it using the conditioning interpretation. For an explanation of why the rule in A level is valid, we won't go through every epsilon-delta detail.

Setup: S is the space we are constrained to which may or may not contain constraints other than only the total. However, the expected table WILL ALWAYS SATISFY THESE CONSTRAINTS as we will assume and use that the distance from the expected table and the center of the lattice will be equal. If y is in the standardized world, then for any space A contained in S and the central region (in fact we care about the squared distance so we only need A to be a circle or sphere, so there is no issue with A being some weird fractal region or anything like that), if Λ_n is the set of lattice points in the standardized world in our lattice that we are interested in constraining to, and R is the central region in the space we are constraining to, then we have the following identity by claims 1, 2 and 3:

$$P(y \in A | y \in S) = \frac{\sum_{z \in \Lambda_n \cap A} P(z)}{\sum_{z \in \Lambda_n \cap S} P(z)} = \frac{\sum_{z \in \Lambda_n \cap A \cap R} P(z) + o(1)}{\sum_{z \in \Lambda_n \cap R} P(z) + o(1)}$$

Since the probability of being in the central region R tends to 1 we have the identity above and furthermore $P(z)$ is proportional to $e^{-\frac{1}{2}|z|^2}$ where I mean the length of z in the standardized world.

In particular by the 3 claims from earlier we have $P(y \in A | y \in S) = \frac{\sum_{z \in A} e^{-\frac{1}{2}(L^2)}}{\sum_{z \in S} e^{-\frac{1}{2}(L^2)}} (1 + o(1))$ (In particular claim 3 allows restriction to lattices like that to prevent any weirdness with the lattices getting smaller in probability making the probability given we are in a lattice getting larger) which we hope to show is

$$\frac{\int_S \text{In radius of A } e^{-\frac{1}{2}L^2}}{\int_S e^{-\frac{1}{2}L^2}} \rightarrow P(\text{A standard normal with the right dimension count is in A} | \text{In S})$$

as then we will have a universal cumulative distribution for this distance A which gives the chi squared distribution when squared. We will simply show that the sums approach the integrals as n goes to infinity then we will be done (since the integrals are non-zero for n large enough).

Start with the situation where all constraints are imposed, then add even more until only a single point is allowed. Now imagine removing the constraints one by one until we are back to where we started.

Each time we remove one constraint, we gain a new freedom of movement: from any allowed point, we can now move in one new straight direction and still remain within the allowed set. Crucially, this direction is genuinely new - it is not just a combination of the directions we already had - because if it were, removing the constraint would not have created any new reachable points.

After all constraints have been removed, we are left with a flat space (the slice we are working in), and there is a fixed collection of directions with the following property:

1. Starting from any allowed point, we can reach any other allowed point by moving some whole number of steps forward or backward along each of these directions.
2. None of these directions is redundant: each one adds a new way of moving that cannot be recreated using the others.

Now fix one allowed point P . From P , draw a line segment to the point obtained by moving one step in the first direction. Do the same for each of the other directions. Using these line segments as edges, we obtain a solid shape (a parallelepiped): it consists of all points you can reach from P by moving partway along each direction, but by no more than one step in any of them.

We call this shape a **repeating cell**: it has the property that if we shift it by any whole-number combination of the allowed directions, the shifted copies fill the entire constrained space without gaps or overlaps. Every allowed point lies in exactly one such shifted copy. This has no points in its interior - This is easy to see if you imagine starting from the point and then when you remove the first constraint connect it to the nearest point, when you remove the second one connect this line to the nearest identical line, and so on. So we can actually tile S with repeating cells such that the points in the lattice are exactly the vertices.

Thus, the constrained space can be tiled by identical parallelepipeds built from these directions, and such a repeating cell always exists.

Now recall that we define the riemann integral in higher dimensions as the limit of the sum of the value of the point at one corner of the tile times the volume of the tile. In the standardized world, distances are divided by $\sqrt{n}$ so the volume converges to 0, and for fixed $\delta > 0$ the parallelepipeds are eventually all contained in a ball of radius δ . Since the normal distribution pdf is uniformly continuous inside R (since its derivative is bounded by M so for any ε , $\delta = \frac{\varepsilon}{M}$ is a suitable uniform continuity constant), it means that for any ε we can make the total amount the function changes within each repeating cell as small as we want by making n large enough, and therefore we can make the total integral on our finite region as close as we want to the normal prediction, by making the error from this and part 2 of the claim both sufficiently small.

9.6 Why we combine cells with expected counts under 5

As for why we combine cells with expected values under 5, it's just a rule of thumb, not really a rigorous statement with anything special about 5, at least as far as I know. Essentially the reason is that for the central limit theorem to work, it has to be possible for us to get tails in both directions. If the expected value was, say 1, then the distribution would not possibly be able to be symmetrical without going into the negative values. Also, if p is small (which is usually when we have small expected values), the cell values become well approximated by a poisson distribution, and by the central limit theorem, the poisson distribution is better approximated by a normal the larger the mean is. As for why 5? You can see from this picture that is about when the poisson distribution starts to look like a normal curve.

Figure 9 shows a smooth curve between the probability points of $Po(3, 4, 5, 6)$ on the same graph, demonstrating that around 5 is when it starts to look like a normal distribution.

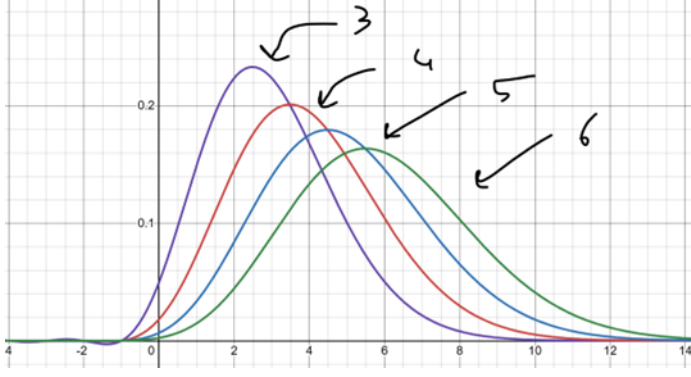


Figure 9

9.7 Parameter estimation in Binomial test case

If there are k cells meaning the binomial has total $k-1$ we would like to say that this results in the standardized vector behaving like a $k-2$ -dimensional normal after conditioning, and indeed that's what this proof used to argue, but since E may not be at the exact middle of this 4-dimensional normal, we need additional justification. Under the null hypothesis, it will be the binomial with some p , but on a generic trial that need not be the observed p , and the expected value induced by this observed p need not coincide with the one that would be at the center of the normal. To see this geometrically, the curve of the possible E vectors for each estimated p need not be orthogonal to the planes of constant estimated p .

In the case of the Poisson test, it actually breaks due to the fact that there are infinitely many outcomes and after grouping we lose information. Therefore we will only consider here the Binomial test. For the Poisson test a very crude heuristic argument is that after we prove it for the binomial we note that a Poisson is the limit of a binomial, but this is not rigorous.

By the Wilson Gerald Ford theorem, it suffices to prove the orthogonality property.

First, we must consider how $p(m^*)$ varies with the mean of the binomial. Write θ for the p of the binomial.

Lemma 3. $P(m^*)$ changes by a factor of $o(1)$ if θ changes by a factor of $O(n^{-1/2})$ which it will do with probability $\rightarrow 1$ as $n \rightarrow \infty$.

Proof. From the derivation in Section 9.5, $\log P(m^*) = \frac{1-k}{2} \log(N) - \frac{1}{2} \sum_{i=1}^k \log(q_i^*) + O(1)$. Because $q_i^* = p_i + O(N^{-1})$, analyzing the behavior of the peak probability reduces to looking at the sum of the log-probabilities of the cells: $S(\theta) = \sum_{i=1}^k \log(p_i(\theta))$, plus possibly a constant that is fixed if we fix N . For the binomial model, the probability of falling into the i -th cell (outcomes 0 to $k-1$) is $p_i(\theta) = \binom{k-1}{i-1} \theta^{i-1} (1-\theta)^{k-i}$. Summing the natural logarithm over all k cells yields:

$$S(\theta) = \sum_{i=1}^k \left[\log \binom{k-1}{i-1} + (i-1) \log(\theta) + (k-i) \log(1-\theta) \right]$$

Since the sum of the first $k-1$ integers is $\frac{k(k-1)}{2}$, this simplifies to:

$$S(\theta) = C_k + \frac{k(k-1)}{2} \log(\theta) + \frac{k(k-1)}{2} \log(1-\theta)$$

where C_k is a constant dependent only on the fixed number of cells k . Crucially, $S(\theta)$ is entirely independent of the total sample size N . In the feasible region, the maximum likelihood estimator $\hat{\theta}$ differs from the true parameter by $O(N^{-1/2})$. Because $S(\theta)$ is a smooth, differentiable function with respect to θ (provided $\theta \notin \{0,1\}$), shifting θ by a distance of $O(N^{-1/2})$ changes the value of $S(\theta)$ by at most $O(N^{-1/2})$. Since $\log P(m^*)$ depends on θ only through this $S(\theta)$ term, $\log P(m^*)$ remains constant up to an $O(N^{-1/2})$ error term. Exponentiating this shows that $P(m^*)$ is scaled by a factor of $1 + O(N^{-1/2})$, meaning it is asymptotically constant as $N \rightarrow \infty$. This smoothness verifies that variations in θ do not distort the probability weights of the central peak.

□

Given this, the underlying geometry of the proof from earlier shows that picking θ to maximize $\frac{P(m)}{P(m^*)}$ approximately coincides with picking θ to minimize the χ^2 statistic and lemma 3 implies that this (approximately) coincides with picking

θ to maximize $P(m)$, as N gets large. Therefore, we would like to show that this coincides with the weighted sum. Once we do this, this will imply that under the hypothesis that it is a binomial we will be looking at a $k-1$ -dimensional normal and projecting out a coordinate so we have $k-2$ degrees of freedom. We don't actually need the conditioning lemmas from the contingency table case for this, since we are no longer conditioning and instead projecting out a coordinate. So here is how we do that. derivation.

We want to find θ that maximizes the probability of a specific table that we observe under the null hypothesis that we have a binomial with some mean. After taking logs and removing the $n!$ term that is constant in m and re-indexing the vector to start with entry 0 for convenience and letting $k+1$ be the number of cells so that k is the n parameter of the binomial we have

$$L(\theta, O_j) = \sum_{j=0}^k O_j \ln \left(\binom{k}{j} \theta^j (1-\theta)^{k-j} \right)$$

The binomial coefficient is constant with respect to θ so if we differentiate this it will vanish. Write

$$L(\theta, O_j) = C + \sum_{j=0}^k O_j [j \ln(\theta) + (k-j) \ln(1-\theta)]$$

Differentiation wrt θ gives

$$L'(\theta, O_j) = \sum_{j=0}^k O_j \left[\frac{j}{\theta} - \frac{k-j}{1-\theta} \right]$$

Let n be the total number of observations and $S = \sum j O_j$ be the total number of successes across all trials. Substituting these in gives

$$L'(\theta, O_j) = \frac{S}{\theta} - \frac{kN - S}{1-\theta}$$

Setting this to 0 gives $\theta = \frac{S}{kN}$, exactly the parameter we would estimate.

10 Other properties of the chi squared and t distributions

In these proofs we will use the fact that a higher dimensional standard normal has the property that its coordinates are independent AND it is spherically symmetric. In particular, a rotation of this random vector has the same distribution as the original and independent components as well.

Proposition. The formula $(n-1) \frac{S^2}{\sigma^2} \sim \chi_{n-1}^2$ is true for independent identically distributed (iid) normally distributed variables

Proof. From the definitions of S^2 , σ^2 and χ_{n-1}^2 , the result would be clear if we could prove it for the case that you were doing this with a $N(0,1)$ variable, as you would observe that $\frac{S^2}{\sigma^2}$ does not change when we scale or move the variable since s scales with σ and neither change by shifting.

Here is why it works for an $N(0,1)$ variable: If I have a bunch of independent random variables, then we can use the same trick of using rotational symmetry of the normal distribution.

Let's define

$$Z_i = \frac{X_i - \mu}{\sigma} \sim N(0, 1)$$

so $X_i = \mu + \sigma Z_i$

We have

$$\sum_{i=1}^n (X_i - \bar{X})^2 = \sigma^2 \sum_{i=1}^n (Z_i - \bar{Z})^2$$

Where $\bar{Z} = \frac{1}{n} \sum_{i=1}^n Z_i$.

Therefore,

$$(n-1) \frac{S^2}{\sigma^2} = \sum_{i=1}^n (Z_i - \bar{Z})^2$$

- The vector $(Z_1, \dots, Z_n)$ is a random point in $\mathbb{R}^n$ distributed as a standard normal vector
- The mean $\bar{Z}$ is the projection of this vector onto the direction $(1, 1, \dots, 1)$
- The quantity $\sum_{i=1}^n (Z_i - \bar{Z})^2$ is the squared length of the component orthogonal to $(1, 1, \dots, 1)$

By rotational symmetry, the “length squared” of this orthogonal bit is distributed as χ_{n-1}^2

This is because the projection of a normal vector onto the plane such that the sample mean is what it is just a normal with one fewer variable in a normal plane by spherical symmetry, and in that case chi squared is clearly the distance squared from the origin of that multivariate normal.

□

Proposition. Suppose we have independent observations from 2 normal distributions X and Y with the same variance and we want to test the difference between their means, and we have $n_X \geq 2$ and $n_Y \geq 2$ observations from each variable respectively. Then if we define

$$s_p^2 = \frac{(n_x - 1) s_{xx} + (n_y - 1) s_{yy}}{n_x + n_y - 2}$$

Then the random variable (if μ_x and μ_y are the means of X and Y respectively)

$$T = \frac{\bar{X} - \bar{Y} - (\mu_x - \mu_y)}{s_p \sqrt{\frac{1}{n_X} + \frac{1}{n_Y}}}$$

has a t-distribution with parameter $n_X + n_Y - 2$

Proof. Define

$$U := \frac{(n_X - 1) s_x^2 + (n_Y - 1) s_y^2}{\sigma^2}$$

Then the projection argument similar to above gives $U \sim \chi_{n_X + n_Y - 2}^2$

Note: People call the parameter of the t distribution the number of degrees of freedom but I’m not really sure how the degrees of freedom relate to the dimension here, so I will merely consider it a parameter.

Note that a t_{n-1} distribution is given by $\frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}}$ in a sample of size n normally distributed variables, where scaling them will affect the numerator and the denominator in the same way and shifting them will not affect the numerator. In the two-sample t-test, we define the following, assuming X and Y have equal variance:

Note that S_p^2 is an unbiased estimate for the variance since it is a linear combination of unbiased estimates for $\text{Var}(X)$ and $\text{Var}(Y)$. We ultimately need to show that T has the same distribution as a $t_{n_X + n_Y - 2}$ variable. Note that we have the following:

$$Z := \frac{\bar{X} - \bar{Y} - (\mu_x - \mu_y)}{\sigma \sqrt{\frac{1}{n_X} + \frac{1}{n_Y}}} \text{ is } N(0, 1)$$

The reason is because the variance of $n_x \bar{X}$, which is the sum of the elements of X, by additivity of variances is the sum of all the variances of the elements of x which is $n_x \sigma^2$. Therefore, since shifting does not affect the variance this is also the variance of $n_x(\bar{X} - \mu)$. Since multiplying a variable by the square of a constant multiplies the variance by that constant, we get that the variance of $\bar{X} - \mu$ must be $\frac{\sigma^2}{n_x}$. Similarly, we have that the variance of $\bar{Y} - \mu$ is $\frac{\sigma^2}{n_y}$. Therefore, we have that the expression above is standardized correctly since the variance of the numerator is $\frac{\sigma^2}{n_x} + \frac{\sigma^2}{n_y}$. We also have the following:

For the same reason as the similar formula we proved above: We simply have to constrain the means of X and Y which reduces the dimension by 2.

Now we have that

$T = \frac{\frac{Z}{U}}{\sqrt{\frac{1}{n_x+n_y-2}}}$ by definition of T. However, this is equal to $\frac{N(0,1)}{\sqrt{\frac{\chi^2_{n_x+n_y-2}}{n_x+n_y-2}}}$ which is equivalent to a $t_{n_x+n_y-2}$ variable in the traditional sense using the relation between S^2 and χ^2 .

Other stuff about t and F distributions are ok because unlike chi-squared distributions we don't need to prove that the critical values are not dependent on something like the probabilities in the table.

□